

SOLVABILITY OF A CLASS OF ABSORPTION PROBLEMS WITH OPTIMAL OPERATIONAL SCALE AND WITHDRAWAL CONTROL

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Abstract

This paper addresses a class of value-maximization problems in storage-type optimal control involving operational scale and withdrawals with absorption. Moving beyond the drifted Brownian motion model, we develop a unified methodological approach under a general linear diffusion framework. We demonstrate the solvability of a broad class of problems by deriving semi-explicit optimal strategies. We further establish conditions for the existence of fully explicit solutions and illustrate solution methods with examples based on models for a variety of scenarios, both where explicit solutions exist and where only semi-explicit solutions are available.

Keywords: Linear diffusion; operational scale; risk control; singular control; withdrawal

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1. Introduction

We study a class of storage-type stochastic control problems with absorption risk, focusing on value maximization. The underlying state process follows a general diffusion, which is controlled in two ways: (i) by adjusting its operational scale through a scaling process that takes values in $[0, 1]$; and (ii) by subtracting a non-decreasing withdrawal process. The controlled process is absorbed once it reaches zero (or below). The objective is to maximize the discounted value of all withdrawals until the absorption time, or indefinitely if absorption never occurs.

This type of stochastic control problem has numerous applications across various fields, including optimal cash flow management in finance (cf. [3, 4, 6, 7, 10, 17, 22, 26]), harvesting planning (cf. [2]), and reservoir management, with the latter two generally involving only the control in (ii). In these contexts, the underlying process typically represents capital or reserves in a company, fluctuating populations in harvesting problems or water levels in a reservoir. In particular, these problems are closely related to the study of optimal dynamic risk exposure control and cash flow management, where the objective is to maximize the expected firm value. Firms strategically manage their risk exposure through business activities and/or

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risk-sharing mechanisms to optimize various objective functions (see, for example, [10, 22, 27, 28]). This application has attracted significant attention over the past few decades, with some of the earliest contributions dating back to [22, 26]. In the standard setting, a firm's liquid reserves are modeled as a diffusion process, where the drift term represents expected profit and the diffusion coefficient reflects risk exposure. The firm controls its risk exposure by adjusting the diffusion coefficient, which simultaneously affects expected profit in the same direction. Extensive literature exists on risk exposure control, including the use of reinsurance as a risk management tool to maximize firm value (see, for example, [3, 4, 6, 7, 9, 10, 12, 13, 17, 22, 26, 29, 31]).

Despite the extensive attention devoted to this type of optimal control problem in the literature, much of the research concerning the controls in both (i) and (ii) above has concentrated on models driven by Brownian motion. Surprisingly, there is still a lack of theoretical studies that address the value-maximization storage-type optimal control of risk exposure/operational scale, and withdrawals in more general and realistic diffusion models that extend beyond the Brownian framework. This paper aims to fill this gap by investigating this class of optimal control problem under a general linear diffusion model, with the goal of deriving explicit theoretical optimality results within this broader setting.

While Brownian motion plays a foundational role in stochastic modeling due to its analytical tractability, it is often too restrictive to accurately capture many real-world phenomena. For example, Brownian motion-based surplus models typically neglect investment or interest income on cash reserves and fail to capture the dependence of returns and volatility on the level of available capital. Although some progress has been made, relatively few studies have explored optimal operational scale and withdrawal strategies within models that move beyond the Brownian motion assumption.

In this paper, we take a step toward addressing this gap by studying a class of optimal operational scale and withdrawal control problems within a general diffusion framework. This class of models includes the classical Brownian motion model as a special case while allowing for more general dynamics, such as state-dependent drift and volatility. We show that these problems can be solved theoretically within this more flexible framework, thereby extending the scope of optimal control results to a wider array of diffusion-based models commonly used in finance and insurance. Our analysis is built upon a general diffusion framework, where both the drift and volatility coefficients are allowed to vary as general functions of the state (or storage) process. This level of generality accommodates a variety of models and enables a unified treatment of optimal control problems across different settings. We demonstrate that, even in this broader context, the optimal control problem is solvable and that the optimal solution to the optimization problem can be obtained through a theoretical approach.

We begin by conjecturing that the positive real line can be divided into three regions: the partial-scale region, the full-scale region, and the withdrawal region. We then theoretically establish the existence of, or implicitly construct, candidate solutions to the Hamilton–Jacobi–Bellman (HJB) equation in each region for any given thresholds. Without requiring explicit closed-form solutions, we derive key theoretical properties of these solutions. A global candidate solution is constructed by smoothly pasting the local solutions across the three regions. By appropriately setting the thresholds, we prove that this global solution satisfies the necessary conditions to be the optimal solution. Our theoretical results hold in a general setting and apply to a broad class of problems. To illustrate the applicability of our approach, we provide examples where explicit solutions can be derived, as well as cases where explicit forms are unavailable within our proposed framework.

This work makes several contributions. We solve a class of value-maximizing optimal control problems involving operational scale/risk exposure, and withdrawals under a generalized diffusion framework. This framework accommodates a broad range of scenarios, including traditional Brownian motion as well as more general diffusion processes. Our approach yields either explicit or semi-explicit solutions. We develop a comprehensive and unified mathematical methodology for addressing these optimization problems. A rigorous and implementable procedure is provided for identifying solutions, whether in cases where closed-form solutions exist or where fully explicit expressions are not obtainable. In addition, we derive necessary and sufficient conditions for the existence of explicit solutions. Finally, to demonstrate the applicability of our results, we present a variety of examples, some rooted in classical settings, and others representing novel scenarios that reflect diverse business contexts for which, to the best of our knowledge, no prior solutions exist.

The remainder of the paper is organized as follows. In Section 2, we introduce the dynamic model and formulate the storage-type value-maximizing operational scale and withdrawals control problem. We divide the analysis into three scenarios based on the behavior of the drift rate: (i) when the drift rate is strictly positive for all levels of storage, (ii) when the drift rate is non-positive at zero storage, and (iii) when the drift becomes negative at some point despite being positive at the origin (zero storage level). These cases are examined in detail in Sections 3–5, where we present the mathematical derivations, optimality results, and semi-explicit forms of the solutions in the general setting. In Section 6, we identify the necessary and sufficient conditions for the availability of fully explicit solutions. Section 7 summarizes the general procedure for solving the optimal control problem. In Section 8, we illustrate the results through several examples, including cases where explicit solutions are available. We also apply the general framework and solution procedure to well-known models of practical interest, such as the Brownian motion model and the Ornstein–Uhlenbeck process. For models where explicit solutions are not tractable, we demonstrate how the same methodology can be used to derive numerical approximations, supported by examples across various scenarios. Section 10 concludes the paper. All technical proofs are provided in the Appendix.

2. Problem formulation

Let $(\Omega, \mathcal{F}, P)$ denote a probability space, where all the random variables involved are defined. Let E represent the expectation associated with P . Define $W = \{W_t\}_{t \geq 0}$ to be a standard Brownian motion and let $\mathcal{F}^W$ denote the augmented complete filtration generated by W . Consider the controlled stochastic process $X_t^{\pi, L}$ governed by the stochastic differential equation (SDE):

$$dX_t^{\pi, L} = \pi_t(\mu(X_{t-}^{\pi, L})dt + \sigma(X_{t-}^{\pi, L})dW_t) - dL_t, \quad t \geq 0, \quad (1)$$

where π_t is $\mathcal{F}_t^W$ -predictable with $\pi_t \in [0, 1]$, L_t is an $\mathcal{F}^W$ -predictable, non-decreasing, and càdlàg (right-continuous with left limits) stochastic process with $L_{0-} = 0$.

We define the hitting time:

$$\tau^{\pi, L} = \inf \{t \geq 0 : X_t^{\pi, L} < 0\}. \quad (2)$$

This hitting time has a natural interpretation in both harvesting planning and reservoir management problems, as it represents the time of extinction and exhaustion, respectively, naturally marking the end of the process. In financial applications, when X_t represents cash reserves, this is the time when reserves hit zero. At this point, we assume the business ceases operations, due to forced liquidation resulting from reserve constraints, as assumed in [20].

We assume the following standard conditions on $\mu(\cdot)$ and $\sigma(\cdot)$: both $\mu(x)$ and $\sigma(x)$ are Lipschitz continuous on $[0, \infty)$ and grow no faster than linearly in x , and $\sigma(\cdot)$ is positive and non-vanishing on $[0, \infty)$. These conditions guarantee that, for an initial state $X_{0-} = x$, and a pair (π, L) (with π_t being $\mathcal{F}^W$ -predictable and L_t being $\mathcal{F}^W$ -predictable, non-decreasing, and càdlàg) there exists a unique strong solution to the SDE (1) on $0 \leq t \leq \tau^{\pi, L}$ (see pp. 37–39 of [21]).

In the above setup, the pair $(\pi, L) := (\{\pi_t\}_{t \geq 0}, \{L_t\}_{t \geq 0})$ is referred to as the *control law*, and $X^{\pi, L} := \{X_t^{\pi, L}\}_{t \geq 0}$ is called the *controlled state process*.

The formulation above describes a controlled stochastic process commonly used in financial modeling, inventory control, and reservoir management. The state variable $X_t^{\pi, L}$ can represent the capital or reserves in a company, inventory levels, or water in a reservoir. The control variable π_t might represent risk exposure or business scale in problems such as consumption, cash withdrawal, or dividends distribution; the proportion of resources allocated to inventory replenishment; or the proportion of inflow directed to storage, among other interpretations. The variable L_t represents the cumulative amount of consumption, cash withdrawal, or dividend payments; non-decreasing adjustments due to external supply, corrections to maintain nonnegativity, or a release process.

We formally define admissible control laws as follows.

Definition 1. (Admissible strategies.) Define $\mathcal{A}$ to be the set of all admissible strategies, (π, L) , that satisfy the following conditions: (a) (π, L) is $\mathcal{F}^W$ -predictable; (b) $\pi_t \in [0, 1]$; (c) L is everywhere right-continuous and has left limits everywhere (càdlàg); and (d) $L_{0-} = 0$, L is non-decreasing and $L_t - L_{t-} \leq X_{t-}^{\pi, L}$.

Item (a) indicates that decisions at any time t depend solely on historical information before time t and item (b) states that the decision-maker can adjust the business scale or exposure ratio anywhere between 0% and 100%. Item (c) implies that if there is a lump sum withdrawal at time t , this amount is treated as a deduction at time $t-$, which reduces the state level immediately before time t . Finally, item (d) specifies that all withdrawals must be non-negative, and the amount of each withdrawal cannot exceed the available quantity at the time of withdrawal. In other words, withdrawals cannot directly result in a deficit.

We quantify the performance of the control law, (π, L) , by the expected total discounted value

$$\mathcal{P}(\pi, L)(x) = \mathbb{E}_x \left[\int_0^{\tau^{\pi, L}} e^{-\delta t} dL_t \right], \quad (3)$$

where $\mathbb{E}_x$ represents the conditional expectation given $X_{0-} = x$:

$$\mathbb{E}_x[\cdot] = \mathbb{E}[\cdot | X_{0-} = x], \quad (4)$$

and $\delta > 0$ is the discount rate used by the decision maker, reflecting their intertemporal preferences.

The optimal control problem. The objective of the control problem is to maximize the *performance functional*

$$\mathcal{P}(\pi, L)(x) = \mathbb{E}_x \left[\int_0^{\tau^{\pi, L}} e^{-\delta t} dL_t \right], \quad (5)$$

over all admissible strategies $(\pi, L) \in \mathcal{A}$, that is,

$$V(x) := \sup_{(\pi, L) \in \mathcal{A}} \mathbb{E}_x \left[\int_0^{\tau^{\pi, L}} e^{-\delta t} dL_t \right], \quad (6)$$

subject to the system dynamics in (1) and the constraints imposed on admissible strategies as outlined in Definition 1. We refer to V as the *value function*. The value function, depending on the application context, can represent the maximum discounted profit, the maximum benefit, or the maximum utility of water release (e.g. for electricity generation or irrigation).

If there exists a control law, say (π^*, L^*) , from the set of admissible strategies that attains the maximal performance, i.e.

$$\mathcal{P}(\pi^*, L^*)(x) = \sup_{(\pi, L) \in \mathcal{A}} \mathbb{E}_x \left[\int_0^{\tau^{\pi, L}} e^{-\delta t} dL_t \right], \quad (7)$$

then (π^*, L^*) is the *optimal control law*.

The model underlying the control problem discussed above is quite general. It includes the widely studied Brownian motion model, where $\mu(\cdot) = \text{constant}$ and $\sigma(\cdot) = \text{constant}$, as well as the less commonly explored extension, the Ornstein–Uhlenbeck process. The latter corresponds to $\mu(x) = c_1 x + c_2$ for two constants c_1 and c_2 , which can represent more realistic dynamics in certain areas, e.g. finance and insurance, where cash reserves generate interest.

Following standard arguments in optimal control (see, for instance, [30]), we derive the following HJB equation:

$$\max \left\{ \max_{a \in [0, 1]} \left(\frac{1}{2} \sigma^2(x) a^2 f''(x) + \mu(x) a f'(x) - \delta f(x) \right), 1 - f'(x) \right\} \leq 0. \quad (8)$$

Intuitively, if the value function is sufficiently smooth, it should satisfy the above HJB equation with a corresponding to the risk exposure control π . Note that given $X_{0-} = 0$, ruin occurs immediately, i.e. $\tau^{\pi, L} = 0$ conditional on $X_{0-} = 0$. As a result, we have

$$V(0) = \sup_{(\pi, L) \in \mathcal{A}} \mathbb{E}_0 \left[\int_0^{\tau^{\pi, L}} e^{-\delta t} dL_t \right] = 0. \quad (9)$$

We conjecture that if V is sufficiently smooth, a heuristic conclusion would be that it solves the HJB equation. This motivates us to study the solution (if any) to the following initial value problem first. Applying a standard method in optimal control by using Itô lemma we can obtain the following verification theorem.

Theorem 1. (Verification theorem.) *If f is a twice continuously differentiable function that satisfies:*

$$\max \left\{ \max_{a \in [0, 1]} \left(\frac{1}{2} \sigma^2(x) a^2 f''(x) + \mu(x) a f'(x) - \delta f(x) \right), 1 - f'(x) \right\} \leq 0, \quad \text{for } x > 0, \quad f(0) = 0, \quad (10)$$

then $f(x) \geq V(x)$ for $x \geq 0$.

In light of the above verification theorem, it is natural to first inquire whether such a function as described in the theorem exists. We attempt to answer this by searching for candidate

solutions to the initial value HJB equation:

$$\max \left\{ \max_{a \in [0, 1]} \left(\frac{1}{2} \sigma^2(x) a^2 f''(x) + \mu(x) a f'(x) - \delta f(x) \right), 1 - f'(x) \right\} = 0, \quad x > 0, \quad (11)$$

$$f(0) = 0. \quad (12)$$

A special case where both $\mu(x)$ and $\sigma(x)$ are constants has been solved in [14], who studied an optimal reinsurance and dividend problem for a Brownian motion with a constant positive drift and a constant diffusion coefficient. This leads to the same mathematical problem as the special case discussed here. Further variations of the problem under the Brownian motion framework (with constant drifts and volatilities, in absence of control) have been considered in [3, 4, 6, 9–11, 15, 17, 18, 25, 26], among others.

We distinguish among three cases, (a) $\mu(x) > 0$ for all $x \geq 0$, (b) $\mu(0) \leq 0$, and (c) $\mu(0) > 0$ and $\mu(x) \leq 0$ for some $x > 0$, and analyze them separately. In our analysis, we also impose the following condition on the drift coefficient function.

Assumption 1. We assume $\mu'(x) < \delta$.

This assumption is introduced for mathematical tractability and is also considered natural in financial modeling, as discussed in [19] and also due to the fact that in the context of profit testing, the subjective discount rate δ used to value future cash flows by the shareholders typically exceeds the growth rate (the force of growth) of a risk business, μ' , by a risk margin.

3. Optimality results when $\mu(x) > 0$ for all $x \geq 0$

We consider the case $\mu(x) > 0$ for all $x \geq 0$ throughout the whole section. We start with constructing a strictly increasing and strictly concave solution to

$$\max_{a \in [0, 1]} \left(\frac{1}{2} a^2 \sigma^2(x) f''(x) + \mu(x) a f'(x) - \delta f(x) \right) = 0, \quad \text{for } x \in (0, x_1], \quad f(0) = 0, \quad (13)$$

where x_1 is a positive constant that is determined later. Suppose the desired solution exists, denoted by $f(x)$ here. Noting that this function, f , is strictly concave, we have $f''(x) < 0$ for $x \in (0, x_1]$ and thus we can see

$$\arg \max_a \left(\frac{1}{2} a^2 \sigma^2(x) f''(x) + \mu(x) a f'(x) - \delta f(x) \right) = -\frac{\mu(x)}{\sigma^2(x)} \frac{f'(x)}{f''(x)}, \quad x \in (0, x_1].$$

Define

$$a_f(x) = -\frac{\mu(x)}{\sigma^2(x)} \frac{f'(x)}{f''(x)}. \quad (14)$$

Suppose there exists an $x_1 > 0$ such that $a_f(x) \in [0, 1]$ for $x \in [0, x_1]$ (the existence of such x_1 will be verified later after we obtain a candidate expression for the desired $f(x)$). Then,

$$\arg \max_{a \in [0, 1]} \left(\frac{1}{2} a^2 \sigma^2(x) f''(x) + \mu(x) a f'(x) - \delta f(x) \right) = a_f(x), \quad x \in (0, x_1]. \quad (15)$$

Thus, by combining (15) with the assumption that f is a solution to (13) we obtain

$$\frac{1}{2} a_f^2(x) \sigma^2(x) f''(x) + \mu(x) a_f(x) f'(x) - \delta f(x) = 0, \quad x \in (0, x_1]. \quad (16)$$

From (14), we have

$$\sigma^2(x)f''(x) = -\frac{\mu(x)f'(x)}{a_f(x)}. \quad (17)$$

Substituting (17) into (16) yields

$$\frac{1}{2}a_f(x)\mu(x)f'(x) - \delta f(x) = 0, \quad x \in (0, x_1]. \quad (18)$$

Differentiating (18) with respect to x gives

$$\frac{1}{2}a_f(x)\mu(x)f''(x) + \left(\frac{1}{2}\mu'(x)a_f(x) + \frac{1}{2}\mu(x)a_f'(x) - \delta\right)f'(x) = 0, \quad x \in (0, x_1]. \quad (19)$$

From (17) we get

$$a_f(x)f''(x) = -\frac{\mu(x)f'(x)}{\sigma^2(x)}.$$

By substituting this into (19) we get

$$\left(-\frac{1}{2}\frac{\mu^2(x)}{\sigma^2(x)} + \frac{1}{2}\mu'(x)a_f(x) + \frac{1}{2}\mu(x)a_f'(x) - \delta\right)f'(x) = 0, \quad x \in (0, x_1]. \quad (20)$$

Recall that f is strictly increasing on $[0, x_1]$ and so we know $f'(x) > 0$ for $x \in (0, x_1]$. Thus, it follows from (20) that

$$-\frac{1}{2}\frac{\mu^2(x)}{\sigma^2(x)} + \frac{1}{2}\mu'(x)a_f(x) + \frac{1}{2}\mu(x)a_f'(x) - \delta = 0, \quad x \in (0, x_1],$$

which implies

$$\mu'(x)a_f(x) + \mu(x)a_f'(x) = 2\delta + \frac{\mu^2(x)}{\sigma^2(x)}, \quad \text{for } 0 < x \leq x_1.$$

That is,

$$\frac{d(\mu(x)a_f(x))}{dx} = 2\delta + \frac{\mu^2(x)}{\sigma^2(x)} \quad \text{for } 0 < x \leq x_1.$$

Solving this differential equation gives

$$a_f(x) = \frac{1}{\mu(x)} \left[\int_0^x \left(2\delta + \frac{\mu^2(y)}{\sigma^2(y)} \right) dy + \mu(0)a_f(0) \right], \quad x \in [0, x_1]. \quad (21)$$

From (17) we know

$$\frac{f''(x)}{f'(x)} = -\frac{\mu(x)}{\sigma^2(x)a_f(x)}.$$

By solving this equation we have

$$f(x) = C_1 - C_2 \int_x^{x_1} \exp\left(\int_z^{x_1} \frac{\mu(y)}{\sigma^2(y)a_f(y)} dy\right) dz, \quad x \in (0, x_1]. \quad (22)$$

Since we want $f(0) = 0$ and so

$$C_1 = C_2 \int_0^{x_1} \exp\left(\int_z^{x_1} \frac{\mu(y)}{\sigma^2(y)a_f(y)} dy\right) dz$$

and thus

$$f(x) = C_2 \int_0^x \exp\left(\int_z^{x_1} \frac{\mu(y)}{\sigma^2(y)a_f(y)} dy\right) dz, \quad x \in (0, x_1]. \quad (23)$$

It follows from (18) that $\frac{1}{2}\mu(0)a_f(0)f'(0+) - \delta f(0) = 0$. Note that $f(0) = 0$, $\mu(0) > 0$, and $f'(0+) > 0$ (as f is strictly increasing on $[0, x_1]$). Therefore, we have $\frac{1}{2}\mu(0)a_f(0)f'(0+) = 0$, and thus

$$a_f(0) = 0, \quad (24)$$

which along with (21) implies

$$a_f(x) = \frac{\int_0^x \left(2\delta + \frac{\mu^2(y)}{\sigma^2(y)}\right) dy}{\mu(x)}, \quad x \in [0, x_1]. \quad (25)$$

Before we proceed with further derivation, let us define two quantities.

Notation 2. Define the following function and quantity:

$$A(x) = \frac{\int_0^x \left(2\delta + \frac{\mu^2(y)}{\sigma^2(y)}\right) dy}{\mu(x)} \quad \text{for } x \geq 0 \quad \text{and} \quad x^* = \inf\{x \geq 0 : A(x) > 1\}. \quad (26)$$

Those quantities possess the following properties.

Lemma 1. Let $A(x)$ and x^* be defined in (26), respectively. The function $A(x)$ is increasing on $[0, x^*]$ and the following hold:

$$0 < x^* < \frac{\mu(0)}{\delta}, \quad (27)$$

$$0 \leq A(x) \leq 1, \quad x \in [0, x^*], \quad (28)$$

$$A(0) = 0, \quad A(x^*) = 1. \quad (29)$$

Now let us resume our analysis on the desired solution to (11) and (12) for $x \in [0, x^*]$. From the above analysis we can see that for the candidate solution f given in (23), $a_f(x) = A(x) \in [0, 1]$ for $0 \leq x \leq x^*$. Thus, we have $x_1 = x^*$. Thus, by (23), (25), and (26) it follows that

$$f(x) = C_2 \int_0^x \exp\left(\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy\right) dz, \quad x \in [0, x^*]. \quad (30)$$

Note that $A(x)$ is increasing and $A(0) = 0$. Thus, $\int_z^{x^*} (\mu(y)/\sigma^2(y)A(y))dy$ is finite for any fixed $z > 0$. However, as z approaches zero, this integral may diverge to infinity. Therefore, ensuring the well-definedness of f in (30), which is equivalent to the well-definedness of the integral

$$\int_0^x \exp\left(\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy\right) dz,$$

becomes crucial. This important aspect has been established in Lemma 9 in the Appendix.

From the previous derivation we know that f defined in (30) satisfies the following equation:

$$\frac{1}{2}\sigma^2(x)A^2(x)f''(x) + \mu(x)A(x)f'(x) - \delta f(x) = 0, \quad x \in (0, x^*]. \quad (31)$$

Furthermore, it follows directly from (22) that for any strictly negative C_2 ,

$$f''(x) < 0, \quad x \in [0, x^*]. \quad (32)$$

We show later that $f(x)$ defined in (30) with C_2 being a positive constant is indeed the strictly increasing and strictly concave solution to (13) on $[0, x^*]$.

We now proceed to construct an increasing and concave solution to (13) for $x \in (x^*, b]$, where b ($b > x^*$) is a constant to be specified later. We conjecture that if f is such solution then $af(x) \geq 1$ for $x \in [x^*, b)$. Based on this conjecture we can show that

$$\max_{a \in [0, 1]} \left(\frac{1}{2}\sigma^2(x)a^2 f''(x) + \mu(x)af'(x) - \delta f(x) \right) = \frac{1}{2}\sigma^2(x)f''(x) + \mu(x)f'(x) - \delta f(x)$$

for $x \in [x^*, b)$. We start by defining two functions.

Notation 3. Let g_1 and g_2 denote the unique solutions to the equation

$$\frac{1}{2}\sigma^2(x)g''(x) + \mu(x)g'(x) - \delta g(x) = 0, \quad x > x^*, \quad (33)$$

respectively, with the following two sets of initial values:

$$g_1(x^*) = 0, \quad g'_1(x^*) = 1, \quad (34)$$

$$g_2(x^*) = 1, \quad g'_2(x^*) = 0. \quad (35)$$

The existence and uniqueness of g_1 and g_2 to the initial value problems are proven in Theorem 5.4.2 of [16]. Due to the continuity of $\mu(\cdot)$ and $\sigma(\cdot)$, it is not hard to see that g'_1 and g'_2 are continuous.

We can see that g_1 and g_2 form a set of fundamental solutions to (33) and therefore, the function defined below is a general solution to (33),

$$g(x) = C_4 g_1(x) + C_5 g_2(x), \quad x \geq x^*, \quad (36)$$

where C_4 and C_5 are constants. Let f be the function specified in (30). We now determine C_2 , C_4 , and C_5 by letting $f(x^*) = g(x^*)$, $f'(x^*) = g'(x^*)$, and $g'(b) = 1$, where b ($b \geq x^*$) is a constant to be specified later. These equations amount to

$$\begin{aligned} C_2 \int_0^{x^*} \exp\left(\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy\right) dz &= C_4 g_1(x^*) + C_5 g_2(x^*) = C_5, \\ C_2 &= C_4 g'_1(x^*) + C_5 g'_2(x^*) = C_4, \quad C_4 g'_1(b) + C_5 g'_2(b) = 1. \end{aligned}$$

We can see that for any $b > x^*$, C_2 , C_4 , and C_5 fulfilling the above equations, denoted by $C_2(b)$, $C_4(b)$, and $C_5(b)$ here, have the following representations:

$$C_5(b) = \frac{\int_0^{x^*} \exp\left(\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy\right) dz}{g'_1(b) + g'_2(b) \int_0^{x^*} \exp\left(\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy\right) dz}, \quad (37)$$

$$C_2(b) = C_4(b) = \frac{1}{g'_1(b) + g'_2(b) \int_0^{x^*} \exp\left(\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy\right) dz}. \quad (38)$$

As discussed earlier,

$$\int_0^{x^*} \exp\left(\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy\right) dz$$

is well defined and finite and so are $C_2(b)$, $C_4(b)$, and $C_5(b)$.

We extend the definitions of $C_2(b)$, $C_4(b)$, and $C_5(b)$ to $b = x^*$ by using the same expressions above. Then,

$$C_2(x^*) = \lim_{b \downarrow x^*} C_2(b), \quad C_4(x^*) = \lim_{b \downarrow x^*} C_4(b), \quad C_5(x^*) = \lim_{b \downarrow x^*} C_5(b). \quad (39)$$

The following properties for g_1 , g_2 , and $C_2(\cdot)$, $C_4(\cdot)$, and $C_5(\cdot)$ can be derived (and are proved in the Appendix), and they prove useful in the subsequent analysis.

Lemma 2. (i) We have $g'_1(x) > 0$ and $g'_2(x) \geq 0$ for $x \geq x^*$.

(ii) Let $C_2(b)$, $C_4(b)$ and $C_5(b)$ be specified in (37)–(39), respectively, for $b \geq x^*$, and $A(x)$ and x^* be defined in (26), respectively. For any $b \geq x^*$, all $C_2(b)$, $C_4(b)$, and $C_5(b)$ are strictly positive.

For any $b \geq x^*$, now write

$$f_b(x) = C_2(b) \int_0^x \exp\left(\int_z^{x^*} \frac{\mu^2(y)}{\sigma^2(y)A(y)} dy\right) dz, \quad x \in [0, x^*], \quad (40)$$

$$g_b(x) = C_4(b)g_1(x) + C_5(b)g_2(x), \quad x \in [x^*, \infty). \quad (41)$$

From the above derivation we know that g_b is a solution to (33) on $[x^*, +\infty)$ and satisfies

$$g'_b(b) = 1, \quad f_b(x^*) = g_b(x^*), \quad \text{and} \quad f'_b(x^* -) = g'_b(x^* +). \quad (42)$$

Thus,

$$g''_b(x^* +) = \frac{2(\delta g_b(x^*) - \mu(x^*)g'_b(x^* +))}{\sigma^2(x^*)} = \frac{2(\delta g_b(x^*) - \mu(x^*)A(x^*)g'_b(x^* +))}{A^2(x^*)\sigma^2(x^*)}, \quad (43)$$

where the last equality follows from noting $A(x^*) = 1$ (by Lemma 1). From previous derivations we know that $f_b(x)$ is a solution to (31) and, thus,

$$f''_b(x^* -) = \frac{2(\delta f_b(x^*) - \mu(x^*)A(x^*)f'_b(x^* -))}{A^2(x^*)\sigma^2(x^*)} = g''_b(x^* +), \quad (44)$$

where the last equality follows from (43).

For any $b > x^*$, define

$$v_b(x) := \begin{cases} f_b(x) := C_2(b) \int_0^x \exp\left(\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy\right) dz & x \in [0, x^*), \\ g_b(x) := C_4(b)g_1(x) + C_5(b)g_2(x) & x \in [x^*, b], \\ v_b(b) + x - b & x \in (b, \infty). \end{cases} \quad (45)$$

Define

$$\mathcal{L}_g(a, x) = \frac{1}{2}\sigma^2(x)a^2g''(x) + \mu(x)ag'(x) - \delta g(x). \quad (46)$$

The following properties for v_b will prove useful.

Theorem 2. Let $A(x)$ and x^* be defined in (26), respectively. Let $v_b(x)$ be defined in (45). For any $b > x^*$:

- (i) the function $v_b(x)$ is continuously differentiable on $[0, +\infty)$ and twice continuously differentiable except at the point, $x = b$, $v'_b(x) > 0$ for $x \geq 0$ and $v'_b(b) = 1$;
- (ii) $v''_b(x) < 0$ and $\max_{a \in [0, 1]} \mathcal{L}_{v_b}(a, x) = \mathcal{L}_{v_b}(A(x), x) = 0$ for $x \in [0, x^*]$;
- (iii) $\mathcal{L}_{v_b}(1, x) = 0$ for $x \in (x^*, b)$;
- (iv) if $v''_b(b-) \leq 0$ then $v''_b(x) \leq 0$ and $\max_{a \in [0, 1]} \mathcal{L}_{v_b}(a, x) = \mathcal{L}_{v_b}(1, x) = 0$ for $x \in (x^*, b)$.

Theorem 3. Let $A(\cdot)$ and x^* be defined as in Notation 2. For $b > x^*$, define the pair: $(\pi^*, L^b) = \{(\pi_t^*, L_t^b); t \geq 0\}$ where

$$\pi_t^* = \begin{cases} A(X_{t-}^{\pi^*, L^b}), & 0 \leq X_{t-}^{\pi^*, L^b} \leq x^*, \\ 1, & X_{t-}^{\pi^*, L^b} > x^*, \end{cases}$$

and L_t^b is the function which causes the controlled process to be reflected downward at the boundary b (that is, L^b is the policy that withdraws all storage in excess of b as and keeps the controlled process reflected at b). Let $v_b(x)$ be defined in (45). For any $b > x^*$, $v_b(x) = \mathcal{P}(\pi^*, L^b)(x)$ for $x \geq 0$.

The proof is omitted here, a similar proof procedure can be found in [29]. Here, X^{π^*, L^b} is a stochastic process reflected at the upper boundary, b , and L^b is a non-decreasing process, subtracting which from X^{π^*, L^b} results in the reflection of X^{π^*, L^b} from b . The existence of the classes of strategy L^b has been confirmed in [24, Section 2] and [10, Section 5].

Notation 4. Let g_1 and g_2 be the fundamental solutions defined in Notation 3. Define

$$\xi(b) = g_1''(b) + g_2''(b) \int_0^{x^*} \exp\left(\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy\right) dz, \quad b \in [x^*, \infty), \quad (47)$$

$$b^* = \inf\{b > x^* : \xi(b) \geq 0\}, \quad \text{and} \quad b^* = +\infty \text{ if } \xi(b) < 0 \text{ for all } b \in [x^*, \infty). \quad (48)$$

If we differentiate the expression for $v_b(x)$ (see (45)) twice we can see that the function $\xi(b)$ basically is

$$v''_b(b-) \left(g'_1(b) + g'_2(b) \int_0^{x^*} \exp\left(\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy\right) dz \right).$$

As has already been shown in Lemma 2 that $g'_1 > 0$ and $g'_2 \geq 0$, then $\xi(b) \geq 0$ is equivalent to $v''_b(b-) \geq 0$. Thus,

$$b^* = \inf\{b > x^* : v''_b(b-) \geq 0\}. \quad (49)$$

Due to the continuity of $\mu(\cdot)$ and $\sigma(\cdot)$, we know g''_1 and g''_2 are also continuous and thus $\xi(b)$ is continuous. Obviously, b^* is greater than or equal to x^* . We first show in the following that b^* is strictly greater than x^* .

Lemma 3. *The following holds: $b^* > x^*$.*

There is a chance that b^* is infinitely large. To explore when b^* is not finite, we first examine the monotonicity and concavity of the two fundamental functions, g_1 and g_2 . We can show that for any solution, g , to $(\sigma^2(x)/2)g''(x) + \mu(x)g'(x) - \delta g(x) = 0$, if for some point x_0 such that $g''(x_0) > 0$ and $g'(x_0) \geq 0$, then

$$g''(x) > 0, \quad x \in (x_0, \infty). \quad (50)$$

This can be proven by a proof by contradiction. Suppose the above claim (50) is not true. Considering the continuity of g'' , the quantity, x_1 , defined by $x_1 =: \inf\{x > x_0 : g''(x) = 0\}$, will be finite with

$$g''(x_1) = 0. \quad (51)$$

We can further see

$$g''(x) > 0, \quad x \in (x_0, x_1), \quad (52)$$

and, as a result,

$$g'(x_1) > g'(x_0) \geq 0. \quad (53)$$

Note $g''(x) = (2/\sigma^2(x))(-\mu(x)g'(x) + \delta g(x))$. Hence, $-\mu(x)g'(x) + \delta g(x) > 0$ for $x \in (x_0, x_1)$, and $-\mu(x_1)g'(x_1) + \delta g(x_1) = 0$. As a result, $(-\mu(x)g'(x) + \delta g(x)) - (-\mu(x_1)g'(x_1) + \delta g(x_1)) > 0$, $x \in (x_0, x_1)$. Dividing both sides of the above equation by $x - x_1$ and then letting $x \uparrow x_1$ lead to $(\delta - \mu'(x_1))g'(x_1) - \mu(x_1)g''(x_1) \leq 0$. Noting $g''(x_1) = 0$ (see (51)), we have $(\delta - \mu'(x_1))g'(x_1) \leq 0$. By noting $g'(x_1) > 0$ (see (53)), we conclude $\mu'(x_1) \geq \delta$, which contradicts the condition that $\mu'(x) < \delta$ for $x > 0$. This implies that (50) is indeed true.

Note

$$g''_2(x^*) = \frac{2(\delta g_2(x^*) - \mu(x^*)g'_2(x^*))}{\sigma^2(x^*)} = \frac{2\delta}{\sigma^2(x^*)} > 0$$

and $g'_2(x^*) = 0$, it follows from (50) that

$$g''_2(b) > 0, \quad \text{for all } b > x^*. \quad (54)$$

It follows from (49) that

$$b^* = +\infty \quad \text{if and only if } g'_1(b) < -g'_2(b) \int_0^{x^*} \exp\left(\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy\right) dz, \quad \text{for all } b > x^*. \quad (55)$$

We now consider the following partition and distinguish two cases:

$$(I) \quad g_1''(b) + g_2''(b) \int_0^{x^*} \exp\left(\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy\right) dz \geq 0 \quad \text{for some } b > x^*;$$

and

$$(II) \quad g_1''(b) + g_2''(b) \int_0^{x^*} \exp\left(\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy\right) dz < 0 \quad \text{for all } b > x^*.$$

3.1. Case (I)

We investigate the case where the following condition holds:

$$g_1''(b) + g_2''(b) \int_0^{x^*} \exp\left(\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy\right) dz \geq 0 \quad \text{for some } b > x^*;$$

From the previous discussion we know that in this case, $x^* < b^* < +\infty$. We show that in this case the function v_{b^*} is a solution to the initial value HJB equation and therefore equals the value function. By using Theorem 2 and the definition of b^* , we can obtain the following results for $v_{b^*}(x)$, which are crucial for deriving the optimality results.

Lemma 4. Assume

$$g_1''(b) + g_2''(b) \int_0^{x^*} \exp\left(\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy\right) dz \geq 0 \quad \text{for some } b > x^*.$$

Let $A(x)$, x^* , and b^* be defined in (26) and (48), respectively. Let $v_{b^*}(x)$ be defined in (45). Then:

- (i) $v_{b^*}(x)$ is twice continuously differentiable, $v'_{b^*}(x) > 0$ for $x \geq 0$, and $v''_{b^*}(x) \leq 0$ for $x \geq 0$;
- (ii) $\max_{a \in [0,1]} \mathcal{L}_{v_{b^*}}(a, x) = \mathcal{L}_{v_{b^*}}(A(x), x) = 0$ for $x \in [0, x^*]$;
- (iii) $\max_{a \in [0,1]} \mathcal{L}_{v_{b^*}}(a, x) = \mathcal{L}_{v_{b^*}}(1, x) = 0$ for $x \in (x^*, b^*]$; and
- (iv) $\max_{a \in [0,1]} \mathcal{L}_{v_{b^*}}(a, x) = \mathcal{L}_{v_{b^*}}(1, x) \leq 0$ for $x \in (b^*, \infty)$.

Consequently, we can derive the following optimality results.

Theorem 4. Let $A(x)$, x^* , b^* , and $C_2(\cdot) - C_5(\cdot)$ be defined in (26), (48), and (37)–(38), respectively. Suppose $b^* < +\infty$ and let $v_{b^*}(x)$ be defined in (45). Then,

$$V(x) = v_{b^*}(x) = \begin{cases} C_2(b^*) \int_0^x \exp\left(\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy\right) dz, & x \in [0, x^*), \\ C_4(b^*)g_1(x) + C_5(b^*)g_2(x), & x \in [x^*, b^*], \\ v_{b^*}(b^*) + x - b^*, & x \in (b^*, \infty), \end{cases} \quad (56)$$

and the optimal strategy is $(\pi^*, L^{b^*}) = \{(\pi_t^*, L_t^{b^*}); t \geq 0\}$, where

$$\pi_t^* = \begin{cases} A(X_{t-}^{\pi^*, L^{b^*}}) & 0 \leq X_{t-}^{\pi^*, L^{b^*}} \leq x^*, \\ 1 & X_{t-}^{\pi^*, L^{b^*}} > x^*, \end{cases}$$

and $L_t^{b^*}$ is the functional which causes the controlled process to be reflected downward at the boundary b^* .

The above theorem proves the existence of an optimal solution and provides a semi-explicit method for finding the optimal solutions. Explicit solutions are available if we can find explicit expressions for g_1 and g_2 , which is feasible if and only if we can find explicit expressions for the solutions to the following second-order ordinary differential equation (ODE): $\frac{1}{2}\sigma^2(x)g''(x) + \mu(x)g'(x) - \delta g(x) = 0$.

If the underlying diffusion models are such that it is not possible to solve $\frac{1}{2}\sigma^2(x)g''(x) + \mu(x)g'(x) - \delta g(x) = 0$ explicitly. Then we can use numerical methods to find g_1 and g_2 and then apply the results in Theorem 4 to compute the optimal solutions and find the optimal strategies.

3.2. Case (II)

We now consider case (II) where the following condition holds:

$$g_1''(b) + g_2''(b) \int_0^{x^*} \exp\left(\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy\right) dz < 0 \quad \text{for all } b > x^*.$$

From the early discussions before Section 3.1 we know that in this case $b^* = +\infty$. We first derive some analytical properties that $v_b(x)$ and some associated quantities possess, which play an important role in solving the optimization problem in such case.

Lemma 5. Consider case (II). Let $A(x)$, x^* , and $v_b(x)$ be defined in (26) and (45), respectively. Then, for any $b > x^*$, the function $v_b(x)$ is continuously differentiable on $[0, +\infty)$ and twice continuously differentiable except at $x = b$, $v_b'(x) \geq 1$ for $x \geq 0$, $v_b'(b-) < 0$, and $\max_{a \in [0, 1]} \mathcal{L}_{v_b}(a, x) = \mathcal{L}_{v_b}(A(x), x) = 0$ for $x \in [0, x^*]$ and $\max_{a \in [0, 1]} \mathcal{L}_{v_b}(a, x) = \mathcal{L}_{v_b}(1, x) = 0$ for $x \in (x^*, b)$.

We show that the quantities $C_2(b)$, $C_4(b)$, and $C_5(b)$ are still finite when b converges to $+\infty$, which then guarantees the existence of the limit of $v_b(x)$ when b converges to $+\infty$.

Lemma 6. If $b^* = +\infty$, then $C_2(b)$, $C_4(b)$, and $C_5(b)$ converge to finite quantities as b goes toward $+\infty$.

As a result, the following optimality results can be derived.

Theorem 5. Consider case (II). Let $A(x)$, x^* , b^* , $C_2(\cdot)$, $C_2(\cdot)$, and $C_5(\cdot)$ be defined in (26), (48), and (37)–(38), respectively. Let $v_b(x)$ be defined same as in (45) for $b > x^*$:

$$v_b(x) = \begin{cases} C_2(b) \int_0^x \exp\left(\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy\right) dz, & x \in [0, x^*), \\ C_4(b)g_1(x) + C_5(b)g_2(x), & x \in [x^*, b], \\ v_b(b) + x - b, & x \in (b, \infty). \end{cases}$$

Then: (i) $V(x) = \lim_{b \uparrow +\infty} v_b(x)$ for $x \geq 0$; and (ii) no optimal strategy exists.

4. Optimality results when $\mu(0) \leq 0$

We show in the following that if $\mu(0) \leq 0$, it is optimal to withdraw the full storage, which leads to immediate absorption.

Theorem 6. Assume $\mu(0) \leq 0$. It holds that $V(x) = x$ for $x \geq 0$, and that the optimal strategy is given by (π^0, L^0) , where $\pi^0(t) \equiv 0$ and L^0 represents the strategy that entails withdrawing all available reserves immediately at time 0.

If this is viewed as a control problem in finance, the above optimality results suggest the following intuition. Recall that $\sigma(\cdot)$ is non-vanishing. If $\mu(0) \leq 0$, then when the state is at 0, the instantaneous return rate is zero or negative, while there remains a non-zero risk. This calls for an immediate withdrawal from the risky business.

5. Optimality results when $\mu(0) > 0$ and $\mu(x) \leq 0$ for some $x > 0$

We now consider the case where $\mu(0) > 0$ and $\mu(x) \leq 0$ for some $x > 0$. Define

$$x_0 = \inf\{x > 0 : \mu(x) \leq 0\}. \quad (57)$$

Then $0 < x_0 < \infty$ and $\mu(x) > 0$ for $0 \leq x < x_0$. Define $A(x)$ and x^* as in (26):

$$A(x) = \frac{\int_0^x (2\delta + \frac{\mu^2(y)}{\sigma^2(y)}) dy}{\mu(x)} \quad \text{for } x \geq 0 \quad \text{and} \quad x^* = \inf\{x \geq 0 : A(x) > 1\}. \quad (58)$$

We can obtain the following properties in this case.

Lemma 7. *The function $A(x)$ is increasing on $[0, x^*]$ and the following hold:*

$$0 < x^* < \min\left(\frac{\mu(0)}{\delta}, x_0\right), \quad (59)$$

$$0 \leq A(x) \leq 1, \quad x \in [0, x^*], \quad (60)$$

$$A(0) = 0, \quad A(x^*) = 1. \quad (61)$$

Define $\xi(b)$ and b^* as previously (in (48)):

$$\xi(b) := g_1''(b) + g_2''(b) \int_0^{x^*} \exp\left(\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy\right) dz, \quad b \in [x^*, \infty), \quad (62)$$

$$b^* = \inf\{b > x^* : \xi(b) \geq 0\} \quad \text{and} \quad b^* = +\infty \text{ if } \xi(b) < 0 \text{ for all } b \in [x^*, \infty), \quad (63)$$

where g_1 and g_2 are the fundamental solutions defined in Notation 3. Then, in this case, we have the following result.

Lemma 8. *The following holds: $x^* < b^* \leq x_0$.*

Let $C_2(\cdot)$, $C_4(\cdot)$, and $C_5(\cdot)$ be defined in the same way as in (37) and (38). Define $v_{b^*}(x)$ as in (45):

$$v_{b^*}(x) = \begin{cases} C_2(b^*) \int_0^x \exp\left(\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy\right) dz, & x \in [0, x^*), \\ C_4(b^*)g_1(x) + C_5(b^*)g_2(x), & x \in [x^*, b^*], \\ v_{b^*}(b^*) + x - b^*, & x \in (b^*, \infty). \end{cases}$$

Then, we can prove the following optimality results.

Theorem 7. *The value function $V(x) = v_{b^*}(x)$ and the optimal strategy is $(\pi^*, L^{b^*}) = \{(\pi_t^*, L_t^{b^*}); t \geq 0\}$, where*

$$\pi_t^* = \begin{cases} A(X_{t-}^{\pi^*, L^{b^*}}) & 0 \leq X_{t-}^{\pi^*, L^{b^*}} \leq x^*, \\ 1 & X_{t-}^{\pi^*, L^{b^*}} > x^*, \end{cases}$$

and $L_t^{b^*}$ is the functional which causes the controlled state process to be reflected downward at the boundary b^* .

We observe that the optimality results remain the same as in the case where $\mu(x) > 0$ for all $x > 0$, provided that b^* is finite (see Theorem 4). In this setting, b^* is indeed always finite and is bounded above by x_0 .

6. Conditions for the availability of explicit solutions

Previously, we established the solvability of the optimal control problem. As shown in Theorem 6, when $\mu(0) \leq 0$, the optimal solution is straightforward: the value function is $V(x) = x$, and the optimal strategy is the full withdrawal of the storage, resulting in immediate absorption.

When $\mu(0) > 0$, the optimality results are divided into two cases:

- Theorems 5 and 4 cover the case where $\mu(x) > 0$ for all $x \geq 0$;
- Theorem 7 addresses the case where $\mu(x) \leq 0$ for some $x > 0$.

From the main theoretical results mentioned previously, we can observe that when $\mu(0) > 0$, explicit expressions for the optimal solutions are available if and only if the function

$$v_b(x) = \begin{cases} C_2(b) \int_0^x \exp\left(\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy\right) dz, & x \in [0, x^*), \\ C_4(b)g_1(x) + C_5(b)g_2(x), & x \in [x^*, b], \\ v_b(b) + x - b, & x \in (b, \infty), \end{cases} \quad (64)$$

has explicit expressions. This is equivalent to requiring that the functions g_1 and g_2 admit explicit expressions, which in turn depends on the availability of explicit solutions of the following second-order ODE, $\frac{1}{2}\sigma^2(x)g''(x) + \mu(x)g'(x) - \delta g(x) = 0$.

Theorem 8. *The conditions for the optimal problem to be solvable explicitly are either:*

- $\mu(0) \leq 0$; or
- $\mu(0) > 0$ and the ODE $\frac{1}{2}\sigma^2(x)g''(x) + \mu(x)g'(x) - \delta g(x) = 0$ has explicit non-trivial solutions.

7. Procedure for solving the optimal control problem

We now summarize the procedure for solving the optimal control problem based on the theoretical results obtained in the previous sections. Regardless of the availability of an explicit expression for the optimal value function, we can always solve the optimization problem by applying the following procedure.

Procedure for finding the optimal solution

Step 1: Calculate $A(x)$ by

$$A(x) = \frac{\int_0^x (2\delta + \frac{\mu^2(y)}{\sigma^2(y)}) dy}{\mu(x)} \quad \text{for } x \geq 0. \quad (65)$$

Step 2: Calculate x^* by searching the first positive root to $A(x) - 1$ (the existence of such is proven in Lemma 1).

Step 3: Find the explicit expressions (or numerical solutions if explicit expressions are not available; see Theorem 8) of g_1 and g_2 , as the unique solutions to the following second-order ODE,

$$\frac{1}{2}\sigma^2(x)g''(x) + \mu(x)g'(x) - \delta g(x) = 0, \quad x > x^*, \quad (66)$$

respectively, with the following two sets of initial values:

$$g_1(x^*) = 0, \quad g'_1(x^*) = 1, \quad (67)$$

$$g_2(x^*) = 1, \quad g'_2(x^*) = 0. \quad (68)$$

Step 4: Check whether

$$\xi(b) = g''_1(b) + g''_2(b) \int_0^{x^*} \exp\left(\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy\right) dz$$

has a finite root on (x^*, ∞) . If yes, go to step 5(a). Otherwise, proceed to step 5(b).

Step 5(a): Calculate b^* as the smallest positive root of $\xi(b)$ and then calculate the value function by

$$V(x) = \begin{cases} C_2(b^*) \int_0^x \exp\left(\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy\right) dz, & x \in [0, x^*), \\ C_4(b^*)g_1(x) + C_5(b^*)g_2(x), & x \in [x^*, b^*], \\ v_{b^*}(b^*) + x - b^*, & x \in (b^*, \infty), \end{cases} \quad (69)$$

where

$$C_5(b^*) = \frac{\int_0^{x^*} \exp\left(\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy\right) dz}{g'_1(b^*) + g'_2(b^*) \int_0^{x^*} \exp\left(\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy\right) dz}, \quad (70)$$

$$C_2(b^*) = C_4(b^*) = \frac{1}{g'_1(b^*) + g'_2(b^*) \int_0^{x^*} \exp\left(\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy\right) dz}. \quad (71)$$

The optimal strategy is $(\pi^*, L^{b^*}) = \{(\pi_t^*, L_t^{b^*}); t \geq 0\}$, where

$$\pi_t^* = \begin{cases} A(X_t^{\pi^*, L^{b^*}}) & 0 \leq X_t^{\pi^*, L^{b^*}} \leq x^*, \\ 1 & X_t^{\pi^*, L^{b^*}} > x^*, \end{cases}$$

and $L_t^{b^*}$ is the functional which causes the controlled state process to be reflected downward at the boundary b^* .

Step 5(b): If b^* is infinitely large, no optimal strategy exists and the supremum value can never be attained. However, the value function is given by the limit $V(x) = \lim_{b \uparrow \infty} V_b(x)$, where $V_b(x)$, defined below, is increasing in b and converges as $b \uparrow \infty$:

$$V_b(x) = \begin{cases} C_2(b) \int_0^x \exp\left(\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy\right) dz, & x \in [0, x^*), \\ C_4(b)g_1(x) + C_5(b)g_2(x), & x \in [x^*, b], \\ v_b(b) + x - b, & x \in (b^*, \infty), \end{cases} \quad (72)$$

with the coefficients given by

$$C_2(b) = C_4(b) = \frac{1}{g'_1(b) + g'_2(b) \int_0^{x^*} \exp\left(\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy\right) dz},$$

$$C_5(b) = \frac{\int_0^{x^*} \exp\left(\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy\right) dz}{g'_1(b) + g'_2(b) \int_0^{x^*} \exp\left(\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy\right) dz}.$$

8. When explicit expressions are available

In this section, we illustrate how to find explicit solutions for models with explicit solutions using specific examples. Specifically, we consider cases where $\mu(\cdot)$ and $\sigma(\cdot)$ assume particular forms, allowing explicit solutions for the second-order ODE in Theorem 8.

8.1. Example 1: the Brownian motion

In this section, we consider a special case where the state process follows a drifted Brownian motion, one of the most widely used models in the area. Assume $\mu(y) \equiv \mu$ with $\mu \geq 0$ and $\sigma(y) \equiv \sigma$ with $\sigma > 0$. In this case, the controlled process has the following dynamics: $dX_t^{\pi, L} = \mu \pi_t dt + \sigma \pi_t dW_t - dL_t$. We distinguish two cases: (a) $\mu = 0$; and (b) $\mu > 0$.

Case (a) $\mu = 0$. When $\mu = 0$, the case is trivial by Theorem 6: the optimal strategy is to distribute all storage, resulting in absorption immediately (in the context of capital structure models, this corresponds to shareholders optimally extracting all remaining assets and declaring bankruptcy). In this case, no risk exposure/business scale adjustment is required at all. That is, $dL_t^* = x$ for $t = 0$ and $dL_t^* \equiv 0$ for $t > 0$, $\tau^{\pi^*, L^*} = 0$, and $V(x) = x$.

Case (b) $\mu > 0$. It follows from (26) that

$$A(x) = \frac{\int_0^x (2\delta + \frac{\mu^2}{\sigma^2}) dy}{\mu} = \left(\frac{2\delta}{\mu} + \frac{\mu}{\sigma^2} \right) x \quad \text{for } x \geq 0,$$

and thus we can solve x^* from (26):

$$x^* = \frac{1}{\frac{2\delta}{\mu} + \frac{\mu}{\sigma^2}} = \frac{\mu \sigma^2}{2\delta \sigma^2 + \mu^2}. \quad (73)$$

Therefore,

$$\int_0^x \exp\left(\int_z^{x^*} \frac{\mu}{\sigma^2 A(y)} dy\right) dz = \int_0^x \exp\left(\int_z^{x^*} \frac{\mu^2}{(2\delta \sigma^2 + \mu^2)y} dy\right) dz = \frac{\mu}{2\delta} \left(\frac{x}{x^*}\right)^{\frac{2\delta \sigma^2}{\mu^2 + 2\delta \sigma^2}}, \quad (74)$$

$$\int_0^{x^*} \exp\left(\int_z^{x^*} \frac{\mu}{\sigma^2 A(y)} dy\right) dz = \frac{\mu^2 + 2\delta \sigma^2}{2\delta \sigma^2} x^* = \frac{\mu}{2\delta}, \quad (75)$$

where the last equalities in (74) and (75) follow from using (73). Let g_1 and g_2 be solutions to the equation, $\frac{1}{2}\sigma^2 g''(x) + \mu g'(x) - \delta g(x) = 0$, respectively, with the following two set of initial

values:

$$\begin{aligned} g_1(x^*) &= 0, & g_1'(x^*) &= 1, \\ g_2(x^*) &= 1, & g_2'(x^*) &= 0. \end{aligned}$$

The two functions g_1 and g_2 can be solved directly from the above equations, and we obtain

$$g_1(x) = \frac{e^{\theta_1(x-x^*)} - e^{\theta_2(x-x^*)}}{\theta_1 - \theta_2}, \quad g_2(x) = \frac{\theta_1 e^{\theta_2(x-x^*)} - \theta_2 e^{\theta_1(x-x^*)}}{\theta_1 - \theta_2},$$

where θ_1 and θ_2 are the positive and negative solutions to $\frac{1}{2}\sigma^2 x^2 + \mu x - \delta = 0$:

$$\theta_1 = \frac{-\mu + \sqrt{\mu^2 + 2\sigma^2\delta}}{\sigma^2}, \quad \theta_2 = \frac{-\mu - \sqrt{\mu^2 + 2\sigma^2\delta}}{\sigma^2}. \quad (76)$$

Then $\xi(b)$ defined in Notation 4 has the following expression:

$$\begin{aligned} \xi(b) &= g_1''(b) + g_2''(b) \int_0^{x^*} \exp\left(\int_z^{x^*} \frac{\mu}{\sigma^2 A(y)} dy\right) dz \\ &= \frac{\theta_1^2 e^{\theta_1(b-x^*)} - \theta_2^2 e^{\theta_2(b-x^*)}}{\theta_1 - \theta_2} + \frac{\theta_1 \theta_2^2 e^{\theta_2(b-x^*)} - \theta_2 \theta_1^2 e^{\theta_1(b-x^*)}}{\theta_1 - \theta_2} \frac{\mu^2 + 2\delta\sigma^2}{2\delta\sigma^2} x^*, \end{aligned} \quad (77)$$

and b^* defined in the same definition is the smallest root on $(x^*, +\infty)$ to

$$\frac{\theta_1^2 e^{\theta_1(b-x^*)} - \theta_2^2 e^{\theta_2(b-x^*)}}{\theta_1 - \theta_2} + \frac{\theta_1 \theta_2^2 e^{\theta_2(b-x^*)} - \theta_2 \theta_1^2 e^{\theta_1(b-x^*)}}{\theta_1 - \theta_2} \frac{\mu}{2\delta} = 0. \quad (78)$$

Solving the above equation we get

$$b^* = \frac{\ln \frac{\theta_2^2(1-\theta_1 \frac{\mu}{2\delta})}{\theta_1^2(1-\theta_2 \frac{\mu}{2\delta})}}{\theta_1 - \theta_2} + x^*, \quad (79)$$

where the last equality follows from using (73). It follows from (37)–(38) that

$$\begin{aligned} C_5(b^*) &= \frac{\int_0^{x^*} \exp\left(\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy\right) dz}{g_1'(b^*) + g_2'(b^*) \int_0^{x^*} \exp\left(\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy\right) dz} \\ &= \frac{\frac{\mu}{2\delta}(\theta_1 - \theta_2)}{\theta_1 e^{\theta_1(b^*-x^*)} - \theta_2 e^{\theta_2(b^*-x^*)} + \theta_1 \theta_2 (e^{\theta_2(b^*-x^*)} - e^{\theta_1(b^*-x^*)}) \frac{\mu}{2\delta}}, \end{aligned} \quad (80)$$

$$\begin{aligned} C_2(b^*) &= C_4(b^*) = \frac{1}{g_1'(b^*) + g_2'(b^*) \int_0^{x^*} e^{\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy} dz} \\ &= \frac{1}{\theta_1 e^{\theta_1(b^*-x^*)} - \theta_2 e^{\theta_2(b^*-x^*)} + \theta_1 \theta_2 (e^{\theta_2(b^*-x^*)} - e^{\theta_1(b^*-x^*)}) \frac{\mu}{2\delta}}. \end{aligned} \quad (81)$$

By Theorem 5 and (74), we know that the value function is

$$V(x) = v_{b^*}(x) = \begin{cases} C_2(b^*) \frac{\mu}{2\delta} \left(\frac{x}{x^*}\right)^{2\delta\sigma^2/(\mu^2+2\delta\sigma^2)}, & x \in [0, x^*), \\ C_4(b^*)g_1(x) + C_5(b^*)g_2(x) = -\frac{\theta_2}{\theta_1(\theta_1 - \theta_2)}e^{\theta_1(x-b^*)} + \frac{\theta_1}{\theta_2(\theta_1 - \theta_2)}e^{\theta_2(x-b^*)}, & x \in [x^*, b^*], \\ v(b^*) + x - b^*, & x \in (b^*, \infty), \end{cases} \quad (82)$$

and the optimal strategy is $(\pi^*, L^{b^*}) = \{(\pi_t^*, L_t^{b^*}); t \geq 0\}$ where

$$\pi_t^* = \begin{cases} A(X_{t-}^{\pi^*}) = \left(\frac{2\delta}{\mu} + \frac{\mu}{\sigma^2}\right) X_{t-}^{\pi^*} & 0 \leq X_{t-}^{\pi^*, L^{b^*}} \leq x^*, \\ 1 & X_{t-}^{\pi^*, L^{b^*}} > x^*, \end{cases}$$

and $L_t^{b^*}$ is the functional which causes the controlled process to be reflected downward at the boundary b^* , which can be determined by solving the smallest solution on $(x^*, +\infty)$ to (78).

The mathematical model in this case is the same as the zero-liability case in [10], which considers a company whose business activities are modeled by a control process $a_t \in [\alpha, \beta]$, $t \geq 0$. In that context, the controlled state process represents the surplus of the company. The control component a_t represents the risk that the company takes at time t , and the component L_t corresponds to the cumulative dividends paid out by the company up to time t . This example is the same as the case with $\alpha = 0$ and $\beta = 1$. Let $V^{\alpha, \beta}$ represent the value function in the above-mentioned reference. When $\alpha = 0$, $\beta = 1$, we can show that

$$V^{\alpha=0, \beta=1}(x) = \begin{cases} \left(\frac{-\theta_2}{(\theta_1 - \theta_2)}e^{\theta_1\Delta} + \frac{\theta_1}{(\theta_1 - \theta_2)}e^{\theta_2\Delta}\right) \frac{\mu}{2\delta} \left(\frac{x}{x^*}\right)^{1-\Gamma}, & x \in [0, x^*), \\ k_1(1)e^{\theta_1(x-x^*)} + k_2(1)e^{\theta_2(x-x^*)}, & x \in [x^*, b^*], \\ k_1(1) + k_2(1) + x - b^*, & x \in (b^*, \infty), \end{cases} \quad (83)$$

where $\gamma = \delta$ (because γ and δ represent the same thing, the discount rate, in [10] and this paper, respectively), $\Gamma = \mu^2/(\mu^2 + 2\sigma^2\gamma)$, $k_1(1) = -\theta_2/\theta_1(\theta_1 - \theta_2)$, $k_2(1) = \theta_1/\theta_2(\theta_1 - \theta_2)$, and

$$\Delta = \frac{\ln\left(-\frac{\theta_2}{\theta_1}\right)}{\theta_1 - \theta_2}.$$

The detailed calculations that lead to (83) are provided in the Appendix.

By using (82) we can obtain

$$\frac{-\theta_2}{(\theta_1 - \theta_2)}e^{\theta_1\Delta} + \frac{\theta_1}{(\theta_1 - \theta_2)}e^{\theta_2\Delta} = C_2(b^*). \quad (84)$$

By noting (84) and then comparing (83) and (82) shows $V^{\alpha=0, \beta=1}(x) = V(x)$. Thus, the value function in the zero liability case with $\alpha = 0$ and $\beta = 1$ in the reference is the same as the value function in this example.

Moreover, the optimal risk policy in the zero-liability case in [10] is

$$a^{\alpha=0, \beta=1}(x) = \frac{\mu^2 + 2\sigma^2\gamma}{\mu\sigma^2}x = A(x)$$

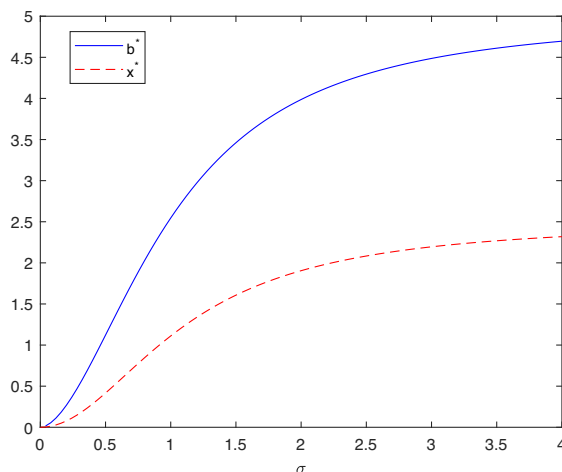


FIGURE 1. The optimal full risk retention threshold, x^* , and the optimal dividend barrier, b^* , as σ varies.

and the optimal dividend strategy in that reference is a barrier strategy with the barrier x_1 , which is exactly the same as the optimal barrier dividend strategy in this example in our paper by noting $x_1 = b^*$ (see (A67) in the Appendix). We can thus conclude that the result in our example is consistent with the result in [10].

Let us now look at some numerical illustrations. Figure 1 plots the optimal full risk retention threshold, x^* , and the optimal dividend barrier, b^* , for various σ , by setting $\mu = 0.5$ and $\delta = 0.1$. Figure 1 illustrates that as the volatility coefficient in the basic process (the process in the absence of control) increases, the threshold for retaining full risk and the dividend barrier also rises. This indicates that the management tends to transfer a higher percentage of risk and retain more surplus when the business becomes riskier. Figure 2 is plotted for various μ by setting $\sigma = 0.5$, and $\delta = 0.1$. In Figure 2, it is evident that when the return rate of the original business (without control), μ , is small, both the risk full retention threshold and the dividend barrier increase as μ increases. Consequently, the management should retain less risk and delay dividend payments until the return rate reaches a certain level. Furthermore, as the return rate continues to rise, the risk full retention threshold and dividend barrier decrease. As a result, the company will retain a higher percentage of risk and pay dividends earlier.

8.2. Example 2: the Ornstein–Uhlenbeck process

We consider an extension to case (I) by including the interest earned by the reserve, which reflects the real situation more closely. We define the drift function as $\mu(x) = c + rx$ and the volatility as $\sigma(y) \equiv \sigma$, where $c, r \geq 0$ and $\sigma > 0$. In this case, the controlled process has the following dynamics:

$$dX_t^{\pi,L} = (c + rX_{t-}^{\pi,L})\pi_t dt + \sigma\pi_t dW_t - dL_t.$$

Such a model is suitable in the context of insurance companies and capital structure modeling. In such context, $X_t^{\pi,L}$ represents the cash reserve/surplus, r represents the constant force of interest earned on the reserve, π_t represents the risk retention ratio, and L_t denotes the cumulative dividend payments. This extension is more realistic compared with the standard Brownian motion, as it acknowledges that reserves or surplus can generate returns and interest over time, and or the drift increases with the scale of business.

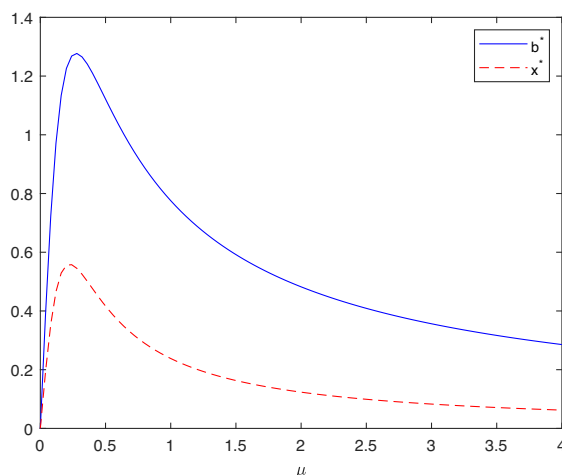


FIGURE 2. The optimal full risk retention threshold, x^* , and the optimal dividend barrier, b^* , as μ varies.

We distinguish among three cases: (a) $c = 0$; (b) $c > 0$ and $r = 0$; and (c) $c > 0$ and $r > 0$.

Case (a) $c = 0$. This case reduces to a trivial case with the take-the-money-and-run strategy to be an optimal strategy. By Theorem 6, the optimal strategy is to distribute all the surplus as dividends and declare bankruptcy immediately. That is, $dL_t^* = x$ for $t = 0$ and $dL_t^* \equiv 0$ for $t > 0$, $\tau^{\pi^*, L^*} = 0$, and $V(x) = x$.

Case (b) $c > 0$ and $r = 0$. This case basically is the Brownian model with a constant drift and it coincides with case (b) in the Brownian case discussed in the previous section.

Case (c) $c > 0$ and $r > 0$. It follows from (26) that

$$A(x) = \frac{\int_0^x \left(2\delta + \frac{(c+ry)^2}{\sigma^2} \right) dy}{c + rx} = \frac{2\delta x + \frac{1}{\sigma^2} \left(c^2 x + crx^2 + \frac{r^2}{3} x^3 \right)}{c + rx}, \quad x \geq 0, \quad (85)$$

and thus we can obtain x^* by finding the smallest positive solution to $A(x) = 1$ (see (26)), which is equivalent to

$$2\delta x + \frac{1}{\sigma^2} \left(c^2 x + crx^2 + \frac{r^2}{3} x^3 \right) - (c + rx) = 0. \quad (86)$$

Write

$$h(x) := 2\delta x + \frac{1}{\sigma^2} \left(c^2 x + crx^2 + \frac{r^2}{3} x^3 \right) - (c + rx).$$

Thus, it amounts to seeking the smallest positive solution to $h(x) = 0$. Note $h(0) = -c < 0$ and $\lim_{x \rightarrow +\infty} h(x) = +\infty$. Thus, there exists at least one positive solution to $h(x) = 0$. Further note that $h''(x) = (1/\sigma^2)(2crx + 2r^2x) > 0$ for $x > 0$. Thus, $h'(x)$ is strictly increasing on $(0, +\infty)$. Hence, if $h'(0) < 0$, $h(x)$ starts with a negative value at $x = 0$, is decreasing until $h'(x)$ increases to 0, and then is increasing. In such case, x^* is the point when $h(x)$ increases to 0, which is unique. Another possibility is that $h'(0) \geq 0$. In this case, $h'(x)$ is strictly positive for all $x > 0$.

Thus, $h(x)$ starts with a negative value at $x=0$ and is always increasing, x^* is the point when $h(x)$ increases to 0, which again is unique. Thus, x^* is the unique positive solution to $h(x)=0$.

It follows from (85) that

$$\begin{aligned}\exp\left(-\int_z^{x^*} \frac{(c+ry)}{\sigma^2 A(y)} dy\right) &= \exp\left(-\int_z^{x^*} \frac{(c+ry)^2}{\sigma^2\left(2\delta y + \frac{1}{\sigma^2}\left(c^2 y + c r y^2 + \frac{r^2}{3} y^3\right)\right)} dy\right) \\ &= \exp\left(-\frac{3}{r^2} \int_z^{x^*} \frac{(c+ry)^2}{(y^2 + a_1 y + a_2)y} dy\right),\end{aligned}$$

where $a_1 = 3c/r$, $a_2 = 3(2\delta\sigma^2 + c^2)/r^2$. Note

$$\begin{aligned}\int_z^{x^*} \frac{(c+ry)^2}{(y^2 + a_1 y + a_2)y} dy &= \int_z^{x^*} \left(\frac{(r^2 - c^2/a_2)y + 2cr - \frac{c^2 a_1}{a_2}}{y^2 + a_1 y + a_2} + \frac{c^2/a_2}{y} \right) dy \\ &= \int_z^{x^*} \left(\frac{(r^2 - c^2/a_2)(y + \frac{a_1}{2}) + 2cr - \frac{c^2 a_1}{2a_2} - \frac{r^2 a_1}{2}}{(y + \frac{a_1}{2})^2 + a_2 - a_1^2/4} + \frac{c^2/a_2}{y} \right) dy \\ &= (r^2 - c^2/a_2) \int_{z+a_1/2}^{x^*+a_1/2} \frac{u}{u^2 + a_2 - a_1^2/4} du + \frac{c^2}{a_2} \ln \frac{x^*}{z} \\ &\quad + \left(2cr - \frac{c^2 a_1}{2a_2} - \frac{r^2 a_1}{2} \right) \int_{z+a_1/2}^{x^*+a_1/2} \frac{1}{u^2 + a_2 - a_1^2/4} du \\ &= \left(\frac{r^2}{2} - \frac{c^2}{2a_2} \right) \ln \frac{(x^*)^2 + a_1 x^* + a_2}{z^2 + a_1 z + a_2} + \frac{c^2}{a_2} \ln \frac{x^*}{z} \\ &\quad + \left(2cr - \frac{c^2 a_1}{2a_2} - \frac{r^2 a_1}{2} \right) \frac{\arctan\left(\frac{x^*+a_1/2}{\sqrt{a_2 - a_1^2/4}}\right) - \arctan\left(\frac{z+a_1/2}{\sqrt{a_2 - a_1^2/4}}\right)}{\sqrt{a_2 - a_1^2/4}}.\end{aligned}$$

Thus,

$$\begin{aligned}\exp\left(\int_z^{x^*} \frac{(c+ry)}{\sigma^2 A(y)} dy\right) &= \left(\frac{x^{*2} + a_1 x^* + a_2}{z^2 + a_1 z + a_2} \right)^{3/2 - 3c^2/2a_2 r^2} \times \left(\frac{x^*}{z} \right)^{3c^2/a_2 r^2} \\ &\quad \times \exp\left(d_1 \left(\arctan\left(\frac{x^* + a_1/2}{\sqrt{a_2 - a_1^2/4}}\right) - \arctan\left(\frac{z + a_1/2}{\sqrt{a_2 - a_1^2/4}}\right) \right)\right),\end{aligned}\tag{87}$$

where

$$d_1 = \frac{6c/r - 3c^2 a_1/(2r^2 a_2) - 3a_1/2}{\sqrt{a_2 - a_1^2/4}}.$$

Then we can get

$$\begin{aligned}
 & \int_0^{x^*} \exp\left(\int_z^{x^*} \frac{c+ry}{\sigma^2 A(y)} dy\right) dz \\
 &= \int_0^{x^*} \left(\frac{x^{*2} + a_1 x^* + a_2}{z^2 + a_1 z + a_2}\right)^{3/2-3c^2/2a_2 r^2} \times \left(\frac{x^*}{z}\right)^{3c^2/a_2 r^2} \\
 &\quad \times \exp\left(d_1 \left(\arctan\left(\frac{x^* + a_1/2}{\sqrt{a_2 - a_1^2/4}}\right) - \arctan\left(\frac{z + a_1/2}{\sqrt{a_2 - a_1^2/4}}\right)\right)\right) dz \\
 &< \int_0^{x^*} \left(\frac{x^{*2} + a_1 x^* + a_2}{a_2}\right)^{3/2-3c^2/2a_2 r^2} \times \left(\frac{x^*}{z}\right)^{3c^2/a_2 r^2} \\
 &\quad \times \exp\left(d_1 \left(\arctan\left(\frac{x^* + a_1/2}{\sqrt{a_2 - a_1^2/4}}\right) - \arctan\left(\frac{z + a_1/2}{\sqrt{a_2 - a_1^2/4}}\right)\right)\right) dz. \quad (88)
 \end{aligned}$$

Note

$$\frac{3c^2}{a_2 r^2} = \frac{3c^2}{\frac{3(2\delta\sigma^2 + c^2)}{r^2} r^2} = \frac{3c^2}{3(2\delta\sigma^2 + c^2)} < 1$$

and so the integral term in (88) is finite.

Let g_1 and g_2 be solutions to the equation, $\frac{1}{2}\sigma^2 g''(x) + (c+rx)g'(x) - \delta g(x) = 0$, respectively, with the following two set of initial values:

$$\begin{aligned}
 g_1(x^*) &= 0, & g_1'(x^*) &= 1; \\
 g_2(x^*) &= 1, & g_2'(x^*) &= 0.
 \end{aligned}$$

The two functions g_1 and g_2 can be solved directly from the above equations and we obtain

$$\begin{aligned}
 g_1(x) &= \frac{-h_2(x^*)}{h_1(x^*)h_2'(x^*) - h_1'(x^*)h_2(x^*)} h_1(x) + \frac{h_1(x^*)}{h_1(x^*)h_2'(x^*) - h_1'(x^*)h_2(x^*)} h_2(x), \\
 g_2(x) &= \frac{-h_2'(x^*)}{h_1'(x^*)h_2(x^*) - h_1(x^*)h_2'(x^*)} h_1(x) + \frac{h_1'(x^*)}{h_1'(x^*)h_2(x^*) - h_1(x^*)h_2'(x^*)} h_2(x),
 \end{aligned}$$

with

$$h_1(x) = U\left(-\frac{\delta}{2r}, \frac{1}{2}, -\frac{(c+rx)^2}{r\sigma^2}\right), \quad h_2(x) = e^{-\frac{(c+rx)^2}{r\sigma^2}} U\left(\frac{1}{2} + \frac{\delta}{2r}, \frac{1}{2}, \frac{(c+rx)^2}{r\sigma^2}\right), \quad (89)$$

$$h_1'(x) = -\frac{\delta(c+rx)}{r\sigma^2} U\left(1 - \frac{\delta}{2r}, \frac{3}{2}, -\frac{(c+rx)^2}{r\sigma^2}\right), \quad (90)$$

$$h_2'(x) = -\frac{c+rx}{\sigma^2} e^{-\frac{(c+rx)^2}{r\sigma^2}} \left(2U\left(\frac{1}{2} + \frac{\delta}{2r}, \frac{1}{2}, \frac{(c+rx)^2}{r\sigma^2}\right) + \left(1 + \frac{\delta}{r}\right) U\left(\frac{3}{2} + \frac{\delta}{2r}, \frac{3}{2}, \frac{(c+rx)^2}{r\sigma^2}\right)\right), \quad (91)$$

where U represents the confluent hypergeometric function of the first type, and the equalities in (90) and (91) are obtained by using the formulae (13.4.21) in [1]. Hence,

$$\begin{aligned} g'_1(x) &= \frac{-h_2(x^*)}{h_1(x^*)h'_2(x^*) - h'_1(x^*)h_2(x^*)} h'_1(x) + \frac{h_1(x^*)}{h_1(x^*)h'_2(x^*) - h'_1(x^*)h_2(x^*)} h'_2(x), \\ g'_2(x) &= \frac{-h'_2(x^*)}{h'_1(x^*)h_2(x^*) - h_1(x^*)h'_2(x^*)} h'_1(x) + \frac{h'_1(x^*)}{h'_1(x^*)h_2(x^*) - h_1(x^*)h'_2(x^*)} h'_2(x), \\ g''_1(x) &= \frac{-h_2(x^*)}{h_1(x^*)h'_2(x^*) - h'_1(x^*)h_2(x^*)} h''_1(x) + \frac{h_1(x^*)}{h_1(x^*)h'_2(x^*) - h'_1(x^*)h_2(x^*)} h''_2(x), \end{aligned} \quad (92)$$

$$g''_2(x) = \frac{-h'_2(x^*)}{h'_1(x^*)h_2(x^*) - h_1(x^*)h'_2(x^*)} h''_1(x) + \frac{h'_1(x^*)}{h'_1(x^*)h_2(x^*) - h_1(x^*)h'_2(x^*)} h''_2(x). \quad (93)$$

Then $\xi(b)$ defined in Notation 4 has the following expression

$$\begin{aligned} \xi(b) &= g''_1(b) + g''_2(b) \int_0^{x^*} \exp\left(\int_z^{x^*} \frac{c+ry}{\sigma^2 A(y)} dy\right) dz \\ &= \frac{-h_2(x^*)}{h_1(x^*)h'_2(x^*) - h'_1(x^*)h_2(x^*)} h''_1(b) + \frac{h_1(x^*)}{h_1(x^*)h'_2(x^*) - h'_1(x^*)h_2(x^*)} h''_2(b) \\ &\quad + \frac{-h'_2(x^*) \int_0^{x^*} \exp\left(\int_z^{x^*} \frac{c+ry}{\sigma^2 A(y)} dy\right) dz}{h'_1(x^*)h_2(x^*) - h_1(x^*)h'_2(x^*)} h''_1(b) + \frac{h'_1(x^*) \int_0^{x^*} \exp\left(\int_z^{x^*} \frac{c+ry}{\sigma^2 A(y)} dy\right) dz}{h'_1(x^*)h_2(x^*) - h_1(x^*)h'_2(x^*)} h''_2(b), \end{aligned} \quad (94)$$

and b^* defined in the same definition is the smallest solution of $\xi(b) = 0$ on $(x^*, +\infty)$. We can show that the above b^* is finite. Note

$$\xi(x^*) = -\frac{2(c+rx^*)}{\sigma^2} + \frac{2\delta}{\sigma^2} \int_0^{x^*} \exp\left(\int_z^{x^*} \frac{c+ry}{\sigma^2 A(y)} dy\right) dz = -\frac{c+rx^*}{\sigma^2} < 0. \quad (95)$$

Thus, it is sufficient to prove $\lim_{b \rightarrow +\infty} \xi(b) > 0$. Before we present the proof, we first define the asymptotic equivalence: any two functions, f and g , are said to be asymptotically equivalent if the limit $\lim_{x \rightarrow \infty} (f(x)/g(x))$ exists and is equal to 1. We denote this as $f(x) \sim g(x)$ as $x \rightarrow \infty$. By using the formulas (13.4.21) and (13.5.2) in [1], we get

$$\begin{aligned} h''_1(b) &\sim \left(-\frac{\delta}{\sigma^2} \left(-\frac{1}{r\sigma^2} \right)^{\delta/2r-1} - \left(\frac{2}{\sigma^2} \right)^2 \frac{\delta}{2r} \left(1 - \frac{\delta}{2r} \right) \left(-\frac{1}{r\sigma^2} \right)^{\delta/2r-2} \right) (c+rb)^{\delta/r-2} \\ &\text{as } b \rightarrow \infty \\ &= \frac{1}{\sigma^4} \left(-\frac{1}{r\sigma^2} \right)^{\delta/2r-2} \left(\left(\frac{\delta}{r} \right)^2 - \frac{\delta}{r} \right) (c+rb)^{\delta/r-2}, \end{aligned} \quad (96)$$

$$h''_2(b) \sim \left(1.5 + \frac{\delta}{2r} \right) \left(\frac{2}{\sigma^2} \right)^2 \left(\frac{1}{r\sigma^2} \right)^{-\delta/2r-1/2} (c+rb)^{-\delta/r+1} e^{-(c+rb)^2/rs^2} \text{ as } b \rightarrow \infty. \quad (97)$$

Thus, it follows from (93) and (94) that

$$g_2''(b) \sim \frac{-h_2'(x^*)}{h_1'(x^*)h_2(x^*) - h_1(x^*)h_2'(x^*)} h_1''(b) \quad \text{as } b \rightarrow \infty, \quad (98)$$

$$\begin{aligned} \xi(b) &\sim \left(h_2(x^*) - h_2'(x^*) \int_0^{x^*} \exp\left(\int_z^{x^*} \frac{c+ry}{\sigma^2 A(y)} dy\right) dz \right) \\ &\quad \times \frac{h_1''(b)}{h_1'(x^*)h_2(x^*) - h_1(x^*)h_2'(x^*)} \quad \text{as } b \rightarrow \infty. \end{aligned} \quad (99)$$

Consequently,

$$\xi(b) \sim \frac{\left(h_2(x^*) - h_2'(x^*) \int_0^{x^*} \exp\left(\int_z^{x^*} \frac{c+ry}{\sigma^2 A(y)} dy\right) dz \right)}{-h_2'(x^*)} g_2''(b) \quad \text{as } b \rightarrow \infty. \quad (100)$$

From (54) we know

$$g_2''(b) > 0, \quad b > x^*. \quad (101)$$

Note the following expression for U (see (13.2.5) in [1]):

$$U(a, b, x) = \frac{1}{\Gamma(a)} \int_0^\infty e^{-xt} t^{a-1} (1+t)^{b-a-1} dt,$$

which implies that for any positive a , $U(a, b, x) > 0$. Hence, from (89) and (91) we obtain $-h_2'(x) > 0$ and $h_2(x) > 0$, which combined with (100) and (100) implies $\xi(b) > 0$ for large b . By noting this, the continuity of $\xi(b)$, and $\xi(x^*) < 0$ (see (95)), we conclude that b^* , the smallest zero of ξ on (x^*, ∞) , exists, that is, $x^* < b^* < +\infty$.

It follows from (37) to (38) that

$$\begin{aligned} C_2(b^*) &= C_4(b^*) = \frac{1}{g_1'(b^*) + g_2'(b^*) \int_0^{x^*} \exp\left(\int_z^{x^*} \frac{c+ry}{\sigma^2 A(y)} dy\right) dz}, \\ C_5(b^*) &= \frac{\int_0^{x^*} \exp\left(\int_z^{x^*} \frac{c+ry}{\sigma^2 A(y)} dy\right) dz}{g_1'(b^*) + g_2'(b^*) \int_0^{x^*} \exp\left(\int_z^{x^*} \frac{c+ry}{\sigma^2 A(y)} dy\right) dz}. \end{aligned}$$

By Theorem 5, we know that the value function is

$$V(x) = \begin{cases} C_1(b) + C_2(b) \int_x^{x^*} \exp\left(\int_z^{x^*} \frac{c+ry}{\sigma^2 A(y)} dy\right) dz & x \in [0, x^*), \\ C_4(b)g_1(x) + C_5(b)g_2(x) & x \in [x^*, b], \\ v(b) + x - b & x \in (b, \infty), \end{cases} \quad (102)$$

where x^* is the unique positive solution to (86), b^* is the smallest solution on $(x^*, +\infty)$ to (94), and the optimal strategy is $(\pi^*, L^{b^*}) = \{(\pi_t^*, L_t^{b^*}); t \geq 0\}$ where

$$\pi_t^* = \begin{cases} A(X_t^{\pi^*}) = \frac{2\delta X_t^{\pi^*} + \frac{1}{\sigma^2} (c^2 X_t^{\pi^*} + cr(X_t^{\pi^*})^2 + \frac{r^2}{3}(X_t^{\pi^*})^3)}{c + rX_t^{\pi^*}}, & 0 \leq X_t^{\pi^*, L^{b^*}} \leq x^*, \\ 1, & X_t^{\pi^*, L^{b^*}} > x^*, \end{cases}$$

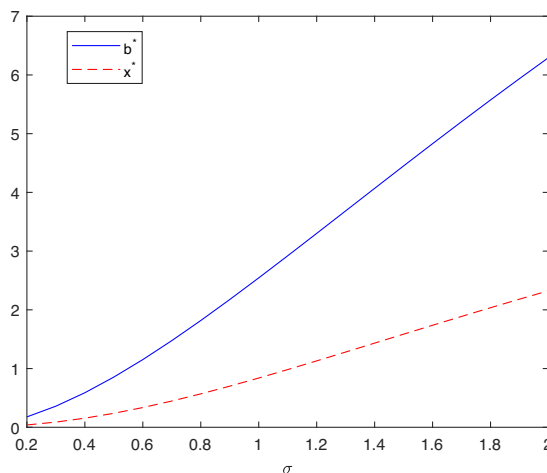


FIGURE 3. The optimal full risk retention threshold, x^* , and the optimal dividend barrier, b^* , as σ varies.

and $L_t^{b^*}$ is the functional which causes the surplus process to be reflected down at the boundary b^* .

Again, let us look at some numerical illustrations. We calculate the optimal threshold for retaining full risk x^* and the optimal dividend barrier b^* by setting $c = 1$, $r = 0.05$, and $\delta = 0.1$ for various σ in Figure 3. In Figure 3, it is evident that with an increase in the volatility coefficient of the uncontrolled process, both the threshold for retaining full risk and the dividend barrier exhibit an upward trend. This observation suggests that as the business becomes riskier, management tends to transfer a higher percentage of risk and retain more surplus. Furthermore, in contrast to the Brownian motion example where the rate of increase in the full risk retention threshold and dividend barrier slows down as σ increases, in this scenario, these thresholds demonstrate a faster rate of increase with larger values of σ . Figure 4 is plotted for various c by setting $r = 0.05$, $\sigma = 1$, and $\delta = 0.1$. We observe a similar pattern for x^* and b^* as c varies to the Brownian motion case. From Figure 4, it is apparent that when the return (drift) rate of the uncontrolled process is small, both the thresholds for retaining full risk and the dividend barrier increase as c rises. This implies that management should assume a lesser degree of risk and postpone dividend payments until the return rate reaches a specific level. Moreover, as the return rate continues to increase, the thresholds for retaining full risk and the dividend barrier decrease, and, consequently, the company will retain a higher proportion of risk and distribute dividends at an earlier stage.

8.3. Example 3

Suppose the drift and volatility coefficients are represented by:

$$\mu(x) = c_1(x + 1), \quad \sigma^2(x) = c_2^2(x + 1)^2$$

with $0 < c_1 < \delta$ and $c_2 > 0$. In this case, the controlled process has the following dynamics:

$$dX_t^{\pi, L} = c_1(rX_{t-}^{\pi, L} + 1)\pi_t dt + c_2(rX_{t-}^{\pi, L} + 1)\pi_t dW_t - dL_t.$$

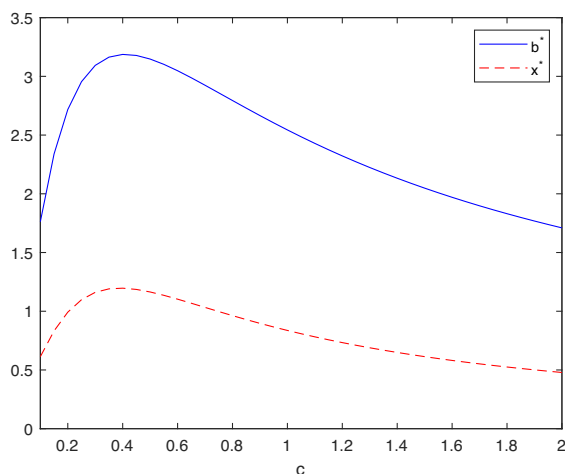


FIGURE 4. The optimal full risk retention threshold, x^* , and the optimal dividend barrier, b^* , as c varies.

Following the procedure presented previously. We first derive

$$A(x) = \frac{\int_0^x \left(2\delta + \frac{\mu^2(y)}{\sigma^2(y)} \right) dy}{\mu(x)} = \frac{(2\delta c_2^2 + c_1^2)x}{c_2^2 c_1 (x+1)}, \quad (103)$$

and thus we can obtain x^* , the smallest positive solution to $A(x) = 1$, which is

$$x^* = \frac{c_2^2 c_1}{2\delta c_2^2 + c_1^2 - c_2^2 c_1}. \quad (104)$$

Note that we can verify $x^* > 0$ because

$$2\delta c_2^2 + c_1^2 - c_2^2 c_1 > 0, \quad (105)$$

which is due to $\delta > c_1$.

For $x \geq x^*$, whether an explicit solution exists is determined by the ODE derived from the HJB equation:

$$\frac{1}{2} c_2^2 (x+1)^2 g''(x) + c_1 (x+1) g'(x) - \delta g(x) = 0.$$

The general solution to this equation is given by

$$g(x) = K_1 (x+1)^{r_1} + K_2 (x+1)^{r_2},$$

where K_1 and K_2 are arbitrary constants, and r_1 and r_2 are the positive and negative roots, respectively, of the characteristic equation:

$$\frac{1}{2} c_2^2 r(r-1) + c_1 r - \delta = 0.$$

The roots are expressed as:

$$r_1 = \frac{-(c_1 - \frac{1}{2} c_2^2) + \sqrt{(c_1 - \frac{1}{2} c_2^2)^2 + 2c_2^2 \delta}}{c_2^2}, \quad r_2 = \frac{-(c_1 - \frac{1}{2} c_2^2) - \sqrt{(c_1 - \frac{1}{2} c_2^2)^2 + 2c_2^2 \delta}}{c_2^2}.$$

Let g_1 and g_2 be solutions to the equation

$$\frac{1}{2}\sigma^2(x)g''(x) + \mu(x)g'(x) - \delta g(x) = 0,$$

corresponding to the following two sets of initial conditions:

$$g_1(x^*) = 0, \quad g_1'(x^*) = 1; \quad g_2(x^*) = 1, \quad g_2'(x^*) = 0.$$

The functions g_1 and g_2 can be explicitly solved from the above equation, yielding the following expressions:

$$g_1(x) = \frac{x^* + 1}{r_1 - r_2} \left(\frac{x + 1}{x^* + 1} \right)^{r_1} + \frac{x^* + 1}{r_2 - r_1} \left(\frac{x + 1}{x^* + 1} \right)^{r_2}, \quad (106)$$

$$g_2(x) = \frac{r_2}{r_2 - r_1} \left(\frac{x + 1}{x^* + 1} \right)^{r_1} + \frac{r_1}{r_1 - r_2} \left(\frac{x + 1}{x^* + 1} \right)^{r_2}. \quad (107)$$

Then $\xi(b)$ defined in Notation 4 has the following expression

$$\begin{aligned} \xi(b) &= g_1''(b) + g_2''(b) \int_0^{x^*} \exp\left(\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy\right) dz \\ &= g_1''(b) + g_2''(b) \int_0^{x^*} \left(\frac{x^*}{z}\right)^{c_1^2/(2\delta c_2^2 + c_1^2)} dz \\ &= \frac{x^* + 1}{r_1 - r_2} \frac{r_1(r_1 - 1)}{(x^* + 1)^{r_1}} (b + 1)^{r_1 - 2} + \frac{x^* + 1}{r_2 - r_1} \frac{r_2(r_2 - 1)}{(x^* + 1)^{r_2}} (b + 1)^{r_2 - 2} \\ &\quad + \left(\frac{r_2}{r_2 - r_1} \frac{r_1(r_1 - 1)}{(x^* + 1)^{r_1}} (b + 1)^{r_1 - 2} + \frac{r_1}{r_1 - r_2} \frac{r_2(r_2 - 1)}{(x^* + 1)^{r_2}} (b + 1)^{r_2 - 2} \right) \left(1 + \frac{c_1^2}{2\delta c_2^2} \right) x^* \\ &= \frac{r_1(r_1 - 1)}{r_1 - r_2} \left(x^* + 1 - r_2 \left(1 + \frac{c_1^2}{2\delta c_2^2} \right) x^* \right) \frac{(b + 1)^{r_1 - 2}}{(x^* + 1)^{r_1}} \\ &\quad - \frac{r_2(r_2 - 1)}{r_1 - r_2} \left(x^* + 1 - r_1 \left(1 + \frac{c_1^2}{2\delta c_2^2} \right) x^* \right) \frac{(b + 1)^{r_2 - 2}}{(x^* + 1)^{r_2}}. \end{aligned}$$

Recall that b^* is the smallest root of $\xi(b)$ on the interval $(x^*, +\infty)$ and, thus, we can solve

$$b^* = (x^* + 1) \cdot \varpi^{1/(r_1 - r_2)} - 1, \quad (108)$$

where

$$\varpi = \frac{r_2(r_2 - 1) \left(x^* + 1 - r_1 \left(1 + \frac{c_1^2}{2\delta c_2^2} \right) x^* \right)}{r_1(r_1 - 1) \left(x^* + 1 - r_2 \left(1 + \frac{c_1^2}{2\delta c_2^2} \right) x^* \right)}. \quad (109)$$

We can indeed easily verify that $b^* > x^*$.

We can calculate

$$\begin{aligned}
 g'_1(b^*) + g'_2(b^*) \int_0^{x^*} e^{\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy} dz &= \frac{r_1}{r_1 - r_2} \left(1 - \frac{r_2}{x^* + 1} \left(1 + \frac{c_1^2}{2\delta c_2^2} \right) x^* \right) \varpi^{(r_1-1)/(r_1-r_2)} \\
 &\quad + \frac{r_2}{r_1 - r_2} \left(\frac{r_1}{x^* + 1} \left(1 + \frac{c_1^2}{2\delta c_2^2} \right) x^* - 1 \right) \varpi^{(r_2-1)/(r_1-r_2)} \\
 &= \frac{r_1}{r_1 - r_2} \left(1 - \frac{r_2 c_1}{2\delta} \right) \varpi^{(r_1-1)/(r_1-r_2)} \\
 &\quad + \frac{r_2}{r_1 - r_2} \left(\frac{r_1 c_1}{2\delta} - 1 \right) \varpi^{(r_2-1)/(r_1-r_2)} \\
 &= \frac{r_1}{1 - r_2} \left(1 - \frac{r_2 c_1}{2\delta} \right) \varpi^{(r_1-1)/(r_1-r_2)}.
 \end{aligned}$$

Using (37) and (38) we can obtain

$$\begin{aligned}
 C_5(b^*) &= \frac{\int_0^{x^*} \exp\left(\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy\right) dz}{g'_1(b^*) + g'_2(b^*) \int_0^{x^*} \exp\left(\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy\right) dz} = \frac{\left(1 + \frac{c_1^2}{2\delta c_2^2}\right) x^*}{\frac{r_1}{1-r_2} \left(1 - \frac{r_2 c_1}{2\delta}\right) \varpi^{(r_1-1)/(r_1-r_2)}}, \\
 C_2(b^*) = C_4(b^*) &= \frac{1}{g'_1(b^*) + g'_2(b^*) \int_0^{x^*} e^{\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy} dz} = \frac{1}{\frac{r_1}{1-r_2} \left(1 - \frac{r_2 c_1}{2\delta}\right) \varpi^{\frac{r_1-1}{r_1-r_2}}}. \quad (110)
 \end{aligned}$$

Then, the value function (by Theorem 5) is

$$\begin{aligned}
 V(x) &= \begin{cases} C_2(b^*) \left(1 + \frac{c_1^2}{2\delta c_2^2}\right) x^{*c_1^2/(2\delta c_2^2+c_1^2)} x^{2\delta c_2^2/(2\delta c_2^2+c_1^2)}, & x \in [0, x^*), \\ C_4(b^*)g_1(x) + C_5(b^*)g_2(x), & x \in [x^*, b^*), \\ V(b^*) + x - b^*, & x \in (b^*, \infty), \end{cases} \\
 &= \begin{cases} \frac{x^{*c_1^2/(2\delta c_2^2+c_1^2)}}{\frac{r_1}{1-r_2} \left(1 - \frac{r_2 c_1}{2\delta}\right) \varpi^{(r_1-1)/(r_1-r_2)}} \left(1 + \frac{c_1^2}{2\delta c_2^2}\right) x^{2\delta c_2^2/(2\delta c_2^2+c_1^2)}, & x \in [0, x^*), \\ \frac{1}{\frac{r_1}{1-r_2} \left(1 - \frac{r_2 c_1}{2\delta}\right) \varpi^{(r_1-1)/(r_1-r_2)}} \left(1 + \frac{c_1^2}{2\delta c_2^2}\right) x^* \\ \quad + \frac{1}{\frac{r_1}{1-r_2} \left(1 - \frac{r_2 c_1}{2\delta}\right) \varpi^{(r_1-1)/(r_1-r_2)}} g_1(x) + \frac{1}{\frac{r_1}{1-r_2} \left(1 - \frac{r_2 c_1}{2\delta}\right) \varpi^{(r_1-1)/(r_1-r_2)}} g_2(x), & x \in [x^*, b^*), \\ V(b^*) + x - b^*, & x \in (b^*, \infty), \end{cases}
 \end{aligned}$$

where ϖ , x^* , and b^* are provided in (109), (104), and (48), and the expressions for the functions, $g_1(x)$ and $g_2(x)$, are given in (106) and (107).

The optimal strategy is $(\pi^*, L^{b^*}) = \{(\pi_t^*, L_t^{b^*}); t \geq 0\}$ where

$$\pi_t^* = \begin{cases} A(X_{t-}^{\pi^*}) = \frac{(2\delta c_2^2 + c_1^2)X_{t-}^{\pi^*}}{c_2^2 c_1 (X_{t-}^{\pi^*} + 1)} & 0 \leq X_{t-}^{\pi^*, L^{b^*}} \leq x^*, \\ 1 & X_{t-}^{\pi^*, L^{b^*}} > x^*, \end{cases}$$

and $L_t^{b^*}$ is the functional which causes the surplus process to be reflected down at the boundary b^* .

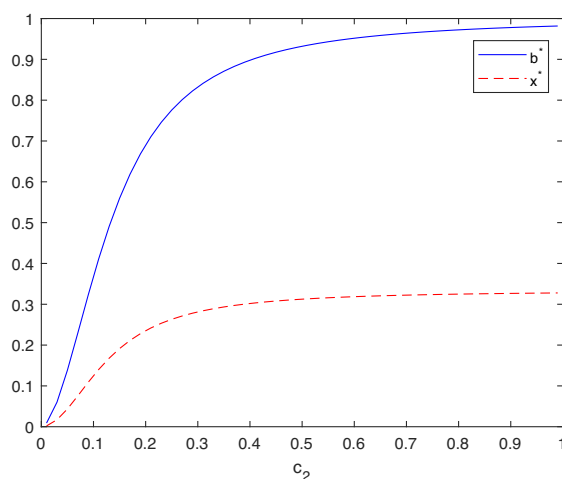


FIGURE 5. The optimal full risk retention threshold, x^* , and the optimal dividend barrier, b^* , as c_2 varies.

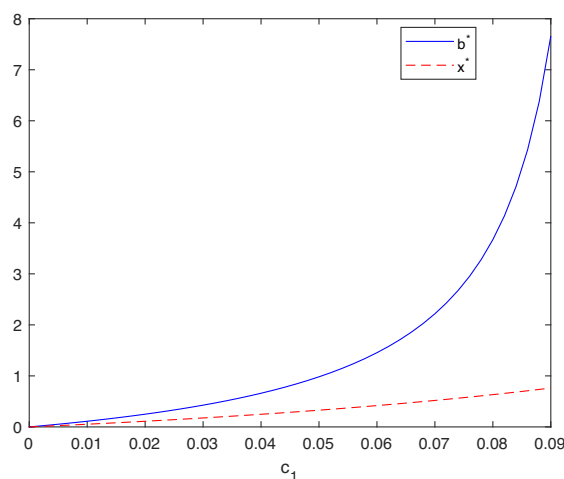


FIGURE 6. The optimal full risk retention threshold, x^* , and the optimal dividend barrier, b^* , as c_1 varies.

Let us now examine some numerical illustrations (see Figures 5 and 6). Let us place this again in the context of insurance risk exposure and capital structure modeling. Figure 5 plots the optimal risk retention threshold, x^* , and the optimal dividend barrier, b^* , for various values of c_2 , with $c_1 = 0.05$ and $\delta = 0.1$. The figure shows that as the volatility coefficient in the uncontrolled process increases, both the full risk retention threshold and the dividend barrier rise, similar to previous cases. Figure 6 illustrates results for various values of c_1 , representing the return/interest rate on the surplus, with $c_2 = 1$ and $\delta = 0.1$. As c_1 increases, both x^* and b^* rise, with b^* growing faster than x^* , indicating a stronger preference for retaining surplus as the return rate on surplus improves.

9. When explicit expressions are not available

We have demonstrated how the optimization problem can be solved using the theoretical results obtained previously when explicit solutions are available. While it is beneficial to have explicit results, in most situations, explicit solutions are not available, though solutions do exist. Actually, explicit expressions are not available if the ODE,

$$\frac{1}{2}\sigma^2(x)g''(x) + \mu(x)g'(x) - \delta g(x) = 0,$$

does not have non-trivial explicit expressions. Now, we provide a few examples to illustrate how our results under the general framework can be used to solve problems when explicit expressions are not available.

The general procedure for finding optimal solutions when explicit expressions are not available is similar to the procedure when explicit solutions exist. We still follow steps 1–4 from Section 8, except that in step 4, we seek numerical solutions for g_1 and g_2 instead of explicit expressions. As illustration, we consider several models that capture different business scenarios, focusing on the drift and volatility functions under various assumptions.

9.1. Model 1: Linear volatility

Suppose that the drift and volatility functions are given by

$$\mu(x) = \mu_0 + c_1x, \quad \sigma(x) = \sigma_0 + c_2x$$

where μ_0 , σ_0 , c_1 , and c_2 are positive constants. The controlled process then has the following dynamics: $dX_t^{\pi,L} = (\mu_0 + c_1X_t^{\pi,L})\pi_t dt + (\sigma_0 + c_2X_t^{\pi,L})\pi_t dW_t - dL_t$. Consider the setting of insurance and capital structure modeling. The expected return rate increases with the surplus process $X_t^{\pi,L}$, which is more realistic compared with a constant drift. The parameter c_1 can also be interpreted as the interest rate. In this case, the absolute volatility increases with the scale of the business, while the relative volatility decreases and stabilizes for sufficiently large scales. Such a model suits businesses that experience economies of scale, where the absolute risk (volatility) grows with the size of the business, but the relative risk decreases as the company becomes larger. This is often applicable to large corporations or industries such as manufacturing and retail, where the operational efficiency improves as the company grows, leading to a more stable relative risk profile despite an increase in the overall scale of operations.

We set $\mu_0 = -0.2$, $c_1 = 0.04$, $\delta = 0.1$, $\sigma_0 = 0.5$, and $c_2 = 0.1$. We can obtain

$$A(x) = \frac{\int_0^x \left(2\delta + \frac{(\mu_0 + c_1y)^2}{(\sigma_0 + c_2y)^2} \right) dy}{\mu_0 + c_1y} = \frac{c_1^2}{c_2^2} \left(x + M_1 \ln \left(\frac{x + N_1}{N_1} \right) - 2M_1^2 \left(\frac{1}{(x + N_1)^3} - \frac{1}{N_1^3} \right) \right), \quad (111)$$

where $M_1 = \mu_0/c_1 - \sigma_0/c_2$ and $N_1 = \sigma_0/c_2$. Furthermore, we can calculate $x^* = 0.62497$, and $b^* = 2.29351$.

Let us place this again in the context of insurance risk exposure and capital structure modeling. Let X_t^* represents controlled process under the optimal strategy. The optimal risk strategy is to retain $100A(X_t^*)\%$ of the risk at time t if $X_t^* < 0.62497$ and to retain the entire risk, 100%, when the reserve is at or exceeds 0.62497, and the optimal cash withdrawal strategy is to distribute minimal dividends so that the surplus is maintained at the level 2.29351. Specifically, no withdrawals are made when the surplus is below b^* , and cash is withdrawn continuously

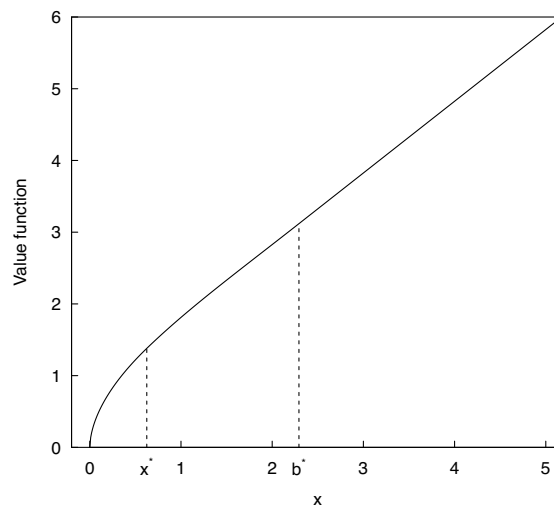


FIGURE 7. The value function under Model 1. Here, the optimal full risk retention threshold is x^* and the optimal dividend barrier is b^* .

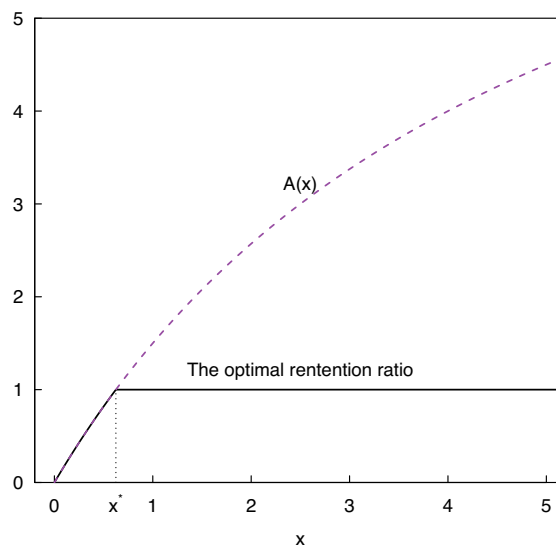


FIGURE 8. The optimal full risk retention ratio under Model 1 with x^* being the full optimal risk retention threshold.

to maintain the controlled process at b^* until it drops below b^* . If the initial reserve exceeds b^* , an amount should be withdrawn immediately to bring the surplus to level b^* . See Figures 7 and 8.

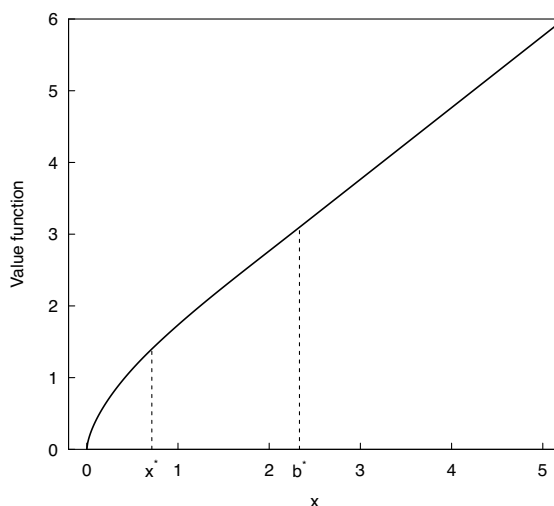


FIGURE 9. The value function under Model 2. Here, the optimal full risk retention threshold is x^* and the optimal dividend barrier is b^* .

9.2. Model 2: The bounded sigmoid-type volatility function

Assume that the drift is linear and the volatility is a bounded sigmoid-type function represented by

$$\mu(x) = \mu_0 + c_1 x, \quad \sigma(x) = \sigma_0 + \frac{c_2}{1 + e^{-x}}, \quad \sigma_0 > 0, c_2 > 0.$$

The controlled process then has the following dynamics:

$$dX_t^{\pi, L} = (\mu_0 + c_1 X_t^{\pi, L}) \pi_t dt + \left(\sigma_0 + \frac{c_2}{1 + \exp(-X_t^{\pi, L})} \right) \pi_t dW_t - dL_t.$$

Here, the absolute volatility $\sigma(x)$ is increasing, while the relative volatility

$$\frac{\sigma(x)}{x} = \frac{\sigma_0 + \frac{c_2}{1+e^{-x}}}{x}$$

is decreasing for large x . As $x \rightarrow \infty$, the relative volatility tends toward zero, reflecting diminishing volatility as the business grows.

We set $\delta = 0.1$, $\mu_0 = 0.2$, $c_1 = 0.04$, $\sigma_0 = 0.5$, and $c_2 = 0.2$. Let us frame this within the context of insurance risk exposure and capital structure modeling again. By following the procedure presented in Section 8, we can compute $x^* = 0.7132967$, $b^* = 2.332054$, the risk retention ratio function $A(x)$, and the value function $V(x)$ (which are depicted as in Figures 9 and 10).

9.3. Model 3: The square root with positive shift volatility function

Now we consider a model where the volatility function is a square root one:

$$\sigma(x) = \sigma_0 + \sqrt{x}, \quad \sigma_0 > 0.$$

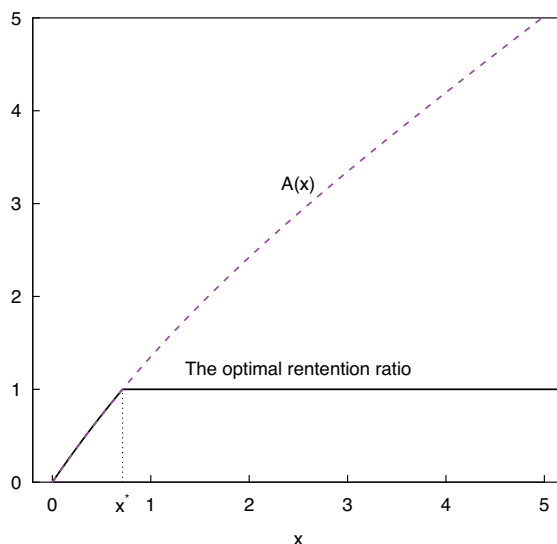


FIGURE 10. The optimal full risk retention ratio under Model 2 with x^* being the optimal full risk retention threshold.

In this model, the absolute volatility $\sigma(x)$ is increasing for $x > 0$, while the relative volatility

$$\frac{\sigma(x)}{x} = \frac{\sigma_0 + \sqrt{x}}{x}$$

decreases for large x , as $\sqrt{x}/x = 1/\sqrt{x}$ diminishes. Again we consider the linear drift function: $\mu(x) = \mu_0 + c_1x$. The controlled process now is

$$dX_t^{\pi,L} = (\mu_0 + c_1X_{t-}^{\pi,L})\pi_t dt + \left(\sigma_0 + \sqrt{X_{t-}^{\pi,L}}\right)\pi_t dW_t - dL_t.$$

We set $\delta = 0.1$, $\mu_0 = 0.2$, $c_1 = 0.04$, $\sigma_0 = 0.5$, and $c_2 = 0.2$. Again, let us situate this within the context of insurance companies and capital structure modeling. By following the procedure presented in Section 8 we can compute $x^* = 0.7068009$ and $b^* = 2.386248$, and further find $A(x)$ and $V(x)$ as in Figures 11 and 12.

9.4. Model 4: The exponential volatility function

Suppose the volatility function is of exponential form:

$$\sigma(x) = \sigma_0 + e^{-\alpha x}, \quad \sigma_0 > 0, \alpha > 0.$$

In this case, the absolute volatility $\sigma(x)$ is decreasing, and the relative volatility

$$\frac{\sigma(x)}{x} = \frac{\sigma_0 + e^{-\alpha x}}{x}$$

is also decreasing for $x > 0$, since $e^{-\alpha x}/x$ decreases as well. Again we consider the linear drift function: $\mu(x) = \mu_0 + c_1x$. The controlled process becomes

$$dX_t^{\pi,L} = (\mu_0 + c_1X_{t-}^{\pi,L})\pi_t dt + \frac{\sigma_0 + \exp(-\alpha X_{t-}^{\pi,L})}{X_{t-}^{\pi,L}}\pi_t dW_t - dL_t.$$

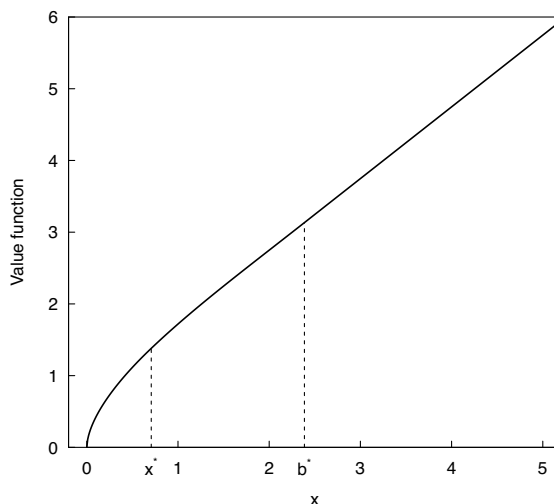


FIGURE 11. The value function under Model 3. Here, the optimal full risk retention threshold is x^* and the optimal dividend barrier is b^* .

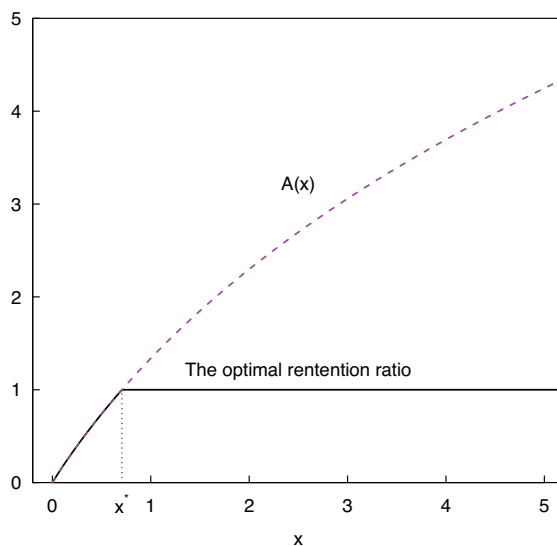


FIGURE 12. The optimal full risk retention ratio under Model 3 with x^* being the full optimal risk retention threshold.

Again, let us place this within the context of insurance companies and capital structure modeling. We set $\delta = 0.1$, $\mu_0 = 0.2$, $c_1 = 0.04$, $\sigma_0 = 0.5$, $c_2 = 0.2$, and $\alpha = 1$. By following the procedure presented in Section 8 we can compute $x^* = 0.7312364$ and $b^* = 2.245842$. The optimal risk retention ratio and the value function are computed and illustrated in Figures 13 and 14.

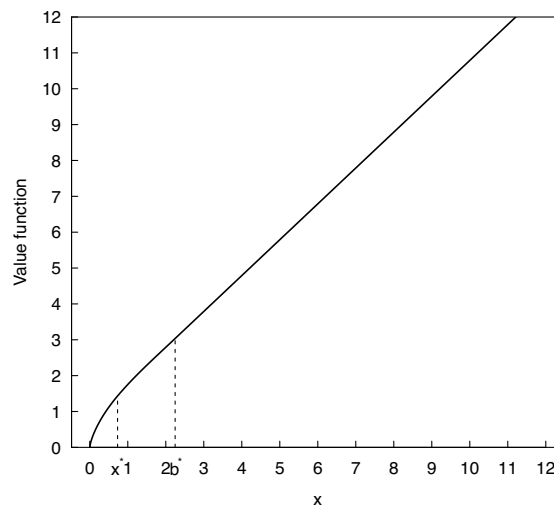


FIGURE 13. The value function under Model 4. Here, the optimal full risk retention threshold is x^* and the optimal dividend barrier is b^* .

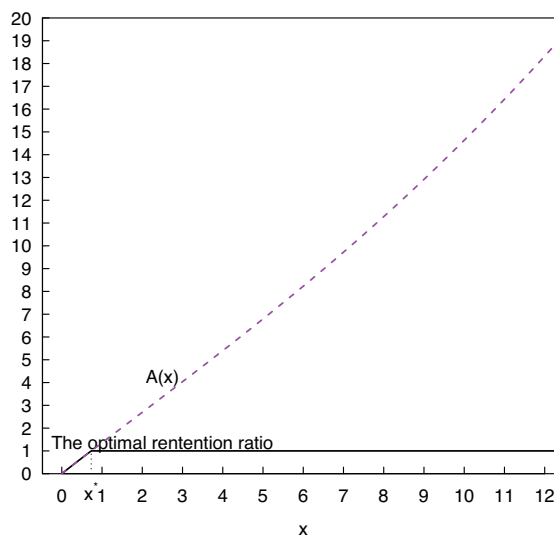


FIGURE 14. The optimal full risk retention ratio under Model 4 with x^* being the optimal risk retention threshold.

10. Conclusion

This paper broadens the study of storage-type value maximization optimal control problems by moving beyond the classical Brownian motion framework to a more general class of linear diffusion processes with state-dependent drift and volatility. This generalization enables a more realistic modeling of the complex dynamics underlying operational scale and withdrawal decisions in finance, insurance, and related fields.

We have developed a unified and rigorous mathematical methodology for determining the value function and optimal strategies that jointly optimize operational scale (or risk exposure)

and withdrawals. The optimization problem is solved within a general linear diffusion framework, and we provide a complete theoretical solution in semi-explicit form. In addition, we identify clear conditions under which fully explicit closed-form solutions are achievable. We present a range of examples involving more realistic models tailored to diverse business scenarios and solve these models explicitly where possible. In cases where closed-form solutions are not available, our semi-explicit approach remains effective, and solutions can be obtained by numerically solving the associated differential equations: demonstrated through a variety of examples and settings.

Our framework is also sufficiently flexible to incorporate additional practical features into the optimization problem, such as bankruptcy penalties, solvency constraints, and additional control variables, which are common in financial applications. Furthermore, the methodology can be extended to address problems involving regular or impulse withdrawal strategies by modifying the candidate solutions accordingly and applying a similar analytical procedure.

Appendix A. Proofs

A.1. Proof of Theorem 1

Let f be the stated solution to (10). Clearly, $f(0) = 0$ and $f'(x) \geq 1$ for $x \geq 0$ and, hence, $f(x) \geq 0$ for $x \geq 0$. Let (π, L) be any admissible strategies pair: $(\pi, L) \in \mathcal{A}$. Note $X_{0-}^{\pi, L} = X_{0-}$. It follows from applying the Itô's formula to $e^{-\delta(\tau^{\pi, L} \wedge t)} f(X_{\tau^{\pi, L} \wedge t}^{\pi, L})$ that

$$\begin{aligned} E_x[e^{-\delta(\tau^{\pi, L} \wedge t)} f(X_{\tau^{\pi, L} \wedge t}^{\pi, L}) - f(X_{0-})] &= E_x \left[\int_0^{\tau^{\pi, L} \wedge t} e^{-\delta s} \left(\frac{1}{2} \sigma^2(X_{s-}^{\pi, L}) \pi_{s-}^2 f''(X_{s-}^{\pi, L}) \right. \right. \\ &\quad \left. \left. + \mu(X_{s-}^{\pi, L}) \pi_{s-} f'(X_{s-}^{\pi, L}) - \delta f(X_{s-}^{\pi, L}) \right) ds \right. \\ &\quad \left. + \int_0^{\tau^{\pi, L} \wedge t} e^{-\delta s} \sigma(X_{s-}^{\pi, L}) \pi_{s-} f(X_{s-}^{\pi, L}) dW_s \right. \\ &\quad \left. - \int_0^{\tau^{\pi, L} \wedge t} e^{-\delta s} f'(X_{s-}^{\pi, L}) d\tilde{L}_s \right. \\ &\quad \left. + \sum_{0 \leq s \leq \tau^{\pi, L} \wedge t} e^{-\delta s} (f(X_s^{\pi, L}) - f(X_{s-}^{\pi, L})) \right] \\ &\leq E_x \left[\int_0^{\tau^{\pi, L} \wedge t} e^{-\delta s} \max_{a \in [0, 1]} \left\{ \frac{1}{2} \sigma^2(X_{s-}^{\pi, L}) a^2 f''(X_{s-}^{\pi, L}) \right. \right. \\ &\quad \left. \left. + \mu(X_{s-}^{\pi, L}) a f'(X_{s-}^{\pi, L}) - \delta f(X_{s-}^{\pi, L}) \right\} ds \right. \\ &\quad \left. + \int_0^{\tau^{\pi, L} \wedge t} e^{-\delta s} \sigma(X_{s-}^{\pi, L}) \pi_{s-} f(X_{s-}^{\pi, L}) dW_s \right. \\ &\quad \left. - \int_0^{\tau^{\pi, L} \wedge t} e^{-\delta s} f'(X_{s-}^{\pi, L}) d\tilde{L}_s \right. \\ &\quad \left. + \sum_{0 \leq s \leq \tau^{\pi, L} \wedge t} e^{-\delta s} (f(X_s^{\pi, L}) - f(X_{s-}^{\pi, L})) \right], \quad (\text{A1}) \end{aligned}$$

where $\tilde{L}$ represents the continuous part of L (that is, $L_s = \tilde{L}_s + L_s - L_{s-}$).

Note that $\int_0^t e^{-\delta s} \sigma(X_{s-}^{\pi,L}) \pi_{s-} f(X_{s-}^{\pi,L}) dW_s$ is a P_x -local martingale (where P_x represents the conditional probability given $X_{0-} = x$), and, thus, there exists a positive sequence t_n with $\lim_{n \rightarrow +\infty} t_n = +\infty$ such that for each n ,

$$E_x \left[\int_0^{\tau^{\pi,L} \wedge t_n} e^{-\delta s} \sigma(X_{s-}^{\pi,L}) \pi_{s-} f(X_{s-}^{\pi,L}) dW_s \right] = 0, \quad x \geq 0. \quad (\text{A2})$$

Note $X_s^{\pi,L} \leq X_{s-}^{\pi,L}$ and $L_s - L_{s-} = X_{s-}^{\pi,L} - X_s^{\pi,L}$. Since $f' \geq 1$ (see (10)), we have

$$-\int_0^{\tau^{\pi,L} \wedge t_n} e^{-\delta s} f'(X_{s-}^{\pi,L}) d\tilde{L}_s \leq -\int_0^{\tau^{\pi,L} \wedge t_n} e^{-\delta s} d\tilde{L}_s, \quad (\text{A3})$$

$$f(X_s^{\pi,L}) - f(X_{s-}^{\pi,L}) \leq X_s^{\pi,L} - X_{s-}^{\pi,L} = L_{s-} - L_s. \quad (\text{A4})$$

By plugging (A2)–(A4) into (A1) and using (10) we arrive at

$$\begin{aligned} E_x \left[e^{-\delta(\tau^{\pi,L} \wedge t_n)} f(X_{\tau^{\pi,L} \wedge t_n}^{\pi,L}) \right] - f(x) &\leq E_x \left[-\int_0^{\tau^{\pi,L} \wedge t_n} e^{-\delta s} d\tilde{L}_s + \sum_{0 \leq s \leq \tau^{\pi,L} \wedge t_n} e^{-\delta s} (L_{s-} - L_s) \right] \\ &= -E_x \left[\int_0^{\tau^{\pi,L} \wedge t_n} e^{-\delta s} dL_s \right]. \end{aligned} \quad (\text{A5})$$

Taking limits $\liminf_{n \uparrow +\infty}$ on both sides leads to

$$\begin{aligned} f(x) &\geq \liminf_{n \uparrow +\infty} E_x \left[e^{-\delta(\tau^{\pi,L} \wedge t_n)} f(X_{\tau^{\pi,L} \wedge t_n}^{\pi,L}) \right] + \liminf_{n \uparrow +\infty} E_x \left[\int_0^{\tau^{\pi,L} \wedge t_n} e^{-\delta s} dL_s \right] \\ &\geq E_x \left[e^{-\delta(\tau^{\pi,L})} f(X_{\tau^{\pi,L}}^{\pi,L}) \right] + E_x \left[\int_0^{\tau^{\pi,L}} e^{-\delta s} dL_s \right], \end{aligned}$$

where the last equality follows from using the Fatou's lemma (noting $f \geq 0$) and the monotone convergence (noting $L_s - L_{s-} \geq 0$). Since $X_{\tau^{\pi,L}}^{\pi,L} = 0$, so $f(X_{\tau^{\pi,L}}^{\pi,L}) = f(0) = 0$ (see (10)). As a result,

$$f(x) \geq E_x \left[\int_0^{\tau^{\pi,L}} e^{-\delta s} dL_s \right] = \mathcal{P}(\pi, L)(x) \quad \text{for } x \geq 0.$$

As the above inequality holds for any admissible pair, (π, L) , we conclude $f(x) \geq \sup_{(\pi, L) \in \mathcal{A}} \mathcal{P}(\pi, L)(x) = V(x)$.

A.2. Proof of Lemma 1

Note $A(0) = a_f(0) = 0$ (by (26) and (24)). Therefore, using the definition of x^* in (26), we can conclude $x^* > 0$. Recall that we are considering the case $\mu(x) > 0$ for $x \geq 0$. From (26) it follows that

$$A(x) \geq \frac{\int_0^x 2\delta dy}{\mu(x)} = \frac{2\delta x}{\mu(x)} \geq \frac{2\delta x}{\mu(0) + \delta x},$$

where the last inequality is due to $\mu'(x) \leq \delta$. Thus, $A(x) \geq 1$ for $x \geq \mu(0)/\delta$. As a result, in view of the definitions of x^* and $A(x)$ in (26), we have $0 < x^* < \mu(0)/\delta$, $0 \leq A(x) \leq 1$ for $0 \leq x \leq x^*$, and $A(x^*) = 1$.

Now we proceed to show that $A(x)$ is increasing. It follows from (26) that

$$A(x)\mu(x) = \int_0^x \left(2\delta + \frac{\mu^2(y)}{\sigma^2(y)} \right) dy \quad \text{for } x \geq 0,$$

and, hence,

$$A'(x)\mu(x) + A(x)\mu'(x) = \left(2\delta + \frac{\mu^2(x)}{\sigma^2(x)} \right) > 2\delta \quad \text{for } x > 0.$$

Further note $0 \leq A(x) \leq 1$ for $x \in [0, x^*]$ and $\mu'(x) < \delta$. Hence, we can obtain $A'(x)\mu(x) + \delta \geq 2\delta$, which leads to $A'(x)\mu(x) \geq \delta$. Recall we are considering the case $\mu(x) > 0$ for $x \in [0, x^*]$. Thus, $A'(x) \geq \delta/\mu(x) > 0$ for $x \in [0, x^*]$ and we conclude that $A(x)$ is increasing on $[0, x^*]$.

Lemma 9. For any $x \geq 0$,

$$\int_0^x \exp\left(\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy\right) dz$$

is well defined and finite.

Proof. To ensure the well-definedness of function f , it is sufficient to show that

$$\int_0^x \exp\left(\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy\right) dz$$

is well defined. After applying (26), we have

$$\int_0^x \exp\left(\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy\right) dz = \int_0^x \exp\left(\int_z^{x^*} \frac{\mu^2(u)/\sigma^2(u)}{\int_0^u (2\delta + \mu^2(y)/\sigma^2(y)) dy} du\right) dz, \quad x \in [0, x^*]. \quad (\text{A6})$$

We show that in the current setup,

$$\int_0^x \exp\left(\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy\right) dz$$

is finite for any $x \in [0, x^*]$. Note that $\sigma(\cdot)$ is non-vanishing and that $\mu(\cdot)$ and $\sigma(\cdot)$ are Lipschitz continuous, and so there exists an $\epsilon_0 > 0$ such that

$$\frac{\mu^2(y)}{\sigma^2(y)} \in \left(\frac{\mu^2(0)}{\sigma^2(0)} - \frac{\delta}{2}, \frac{\mu^2(0)}{\sigma^2(0)} + \frac{\delta}{2} \right) \quad \text{for all } y \in [0, \epsilon_0].$$

As a result,

$$0 \leq \max_{y \in [0, \epsilon_0]} \frac{\mu^2(y)}{\sigma^2(y)} \leq \frac{\mu^2(0)}{\sigma^2(0)} + \frac{\delta}{2}$$

and

$$\min_{y \in [0, \epsilon_0]} \frac{\mu^2(y)}{\sigma^2(y)} \geq \max\left(\frac{\mu^2(0)}{\sigma^2(0)} - \frac{\delta}{2}, 0\right).$$

Therefore,

$$\begin{aligned}
 1 - \frac{\max_{y \in [0, \epsilon_0]} \frac{\mu^2(y)}{\sigma^2(y)}}{2\delta + \min_{y \in [0, \epsilon_0]} \frac{\mu^2(y)}{\sigma^2(y)}} &> 1 - \frac{\frac{\mu^2(0)}{\sigma^2(0)} + \frac{\delta}{2}}{2\delta + \max\left(\frac{\mu^2(0)}{\sigma^2(0)} - \frac{\delta}{2}, 0\right)} \\
 &= \frac{2\delta + \max\left(\frac{\mu^2(0)}{\sigma^2(0)} - \frac{\delta}{2}, 0\right) - \left(\frac{\mu^2(0)}{\sigma^2(0)} + \frac{\delta}{2}\right)}{2\delta + \max\left(\frac{\mu^2(0)}{\sigma^2(0)} - \frac{\delta}{2}, 0\right)} \\
 &\geq \frac{2\delta + \frac{\mu^2(0)}{\sigma^2(0)} - \frac{\delta}{2} - \left(\frac{\mu^2(0)}{\sigma^2(0)} + \frac{\delta}{2}\right)}{2\delta + \max\left(\frac{\mu^2(0)}{\sigma^2(0)} - \frac{\delta}{2}, 0\right)} \\
 &= \frac{\delta}{2\delta + \max\left(\frac{\mu^2(0)}{\sigma^2(0)} - \frac{\delta}{2}, 0\right)} > 0.
 \end{aligned} \tag{A7}$$

Note we can rewrite

$$\begin{aligned}
 &\int_0^x e^{\int_z^{x^*} \frac{\mu^2(u)/\sigma^2(u)}{\int_0^u (2\delta + \mu^2(y)/\sigma^2(y))dy} du} dz \\
 &= \left(\int_0^{\epsilon_0 \wedge x} + \int_{\epsilon_0 \wedge x}^x \right) e^{\int_z^{x^*} \frac{\mu^2(u)/\sigma^2(u)}{\int_0^u (2\delta + \mu^2(y)/\sigma^2(y))dy} du} dz \\
 &\leq \int_0^{\epsilon_0} e^{\int_z^{x^*} \frac{\mu^2(u)/\sigma^2(u)}{\int_0^u (2\delta + \mu^2(y)/\sigma^2(y))dy} du} dz + \int_{\epsilon_0 \wedge x}^x e^{\int_z^{x^*} \frac{\mu^2(u)/\sigma^2(u)}{\int_0^u (2\delta + \mu^2(y)/\sigma^2(y))dy} du} dz \\
 &= \int_0^{\epsilon_0} e^{\left(\int_z^{\epsilon_0 \wedge x^*} + \int_{\epsilon_0 \wedge x^*}^{x^*} \right) \frac{\mu^2(u)/\sigma^2(u)}{\int_0^u (2\delta + \mu^2(y)/\sigma^2(y))dy} du} dz \\
 &\quad + \int_{\epsilon_0 \wedge x}^x e^{\int_z^{x^*} \frac{\mu^2(u)/\sigma^2(u)}{\int_0^u (2\delta + \mu^2(y)/\sigma^2(y))dy} du} dz \\
 &= \left(\int_0^{\epsilon_0} e^{\int_z^{\epsilon_0 \wedge x^*} \frac{\mu^2(u)/\sigma^2(u)}{\int_0^u (2\delta + \mu^2(y)/\sigma^2(y))dy} du} dz \right) \left(e^{\int_{\epsilon_0 \wedge x^*}^{x^*} \frac{\mu^2(u)/\sigma^2(u)}{\int_0^u (2\delta + \mu^2(y)/\sigma^2(y))dy} du} \right) \\
 &\quad + \int_{\epsilon_0 \wedge x}^x e^{\int_z^{x^*} \frac{\mu^2(u)/\sigma^2(u)}{\int_0^u (2\delta + \mu^2(y)/\sigma^2(y))dy} du} dz.
 \end{aligned} \tag{A8}$$

The first term,

$$\int_0^{\epsilon_0} \exp\left(\int_z^{\epsilon_0 \wedge x^*} \frac{\mu^2(u)/\sigma^2(u)}{\int_0^u (2\delta + \mu^2(y)/\sigma^2(y))dy} du\right) dz,$$

is smaller than

$$\int_0^{\epsilon_0} e^{\int_z^{\epsilon_0 \wedge x^*} \frac{\max_{y \in [0, \epsilon_0]} \frac{\mu^2(y)}{\sigma^2(y)}}{\left(2\delta + \min_{y \in [0, \epsilon_0]} \frac{\mu^2(y)}{\sigma^2(y)}\right)u} du} dz = \int_0^{\epsilon_0} \left(\frac{\epsilon_0 \wedge x^*}{z}\right)^{2\delta + \min_{y \in [0, \epsilon_0]} \frac{\mu^2(y)}{\sigma^2(y)}} dz.$$

The latter term is finite by noting

$$\frac{\max_{y \in [0, \epsilon_0]} \frac{\mu^2(y)}{\sigma^2(y)}}{2\delta + \min_{y \in [0, \epsilon_0]} \frac{\mu^2(y)}{\sigma^2(y)}} < 1$$

(see (A7)), which implies that the first term of (A8) is finite. The middle term in (A8) is a finite positive constant by noticing

$$\exp\left(\int_{\epsilon_0 \wedge x^*}^{x^*} \frac{\mu^2(u)/\sigma^2(u)}{\int_0^u (2\delta + \mu^2(y)/\sigma^2(y))dy} du\right) \leq \exp\left(\int_{\epsilon_0 \wedge x^*}^{x^*} \frac{\mu^2(u)/\sigma^2(u)}{\int_0^{\epsilon_0 \wedge x^*} (2\delta + \mu^2(y)/\sigma^2(y))dy} du\right),$$

the positivity of all the terms involved and the smoothness of $\mu(\cdot)$ and $\sigma(\cdot)$. For the same reason, the last term

$$\begin{aligned} & \int_{\epsilon_0 \wedge x}^x \exp\left(\int_z^{x^*} \frac{\mu^2(u)/\sigma^2(u)}{\int_0^u (2\delta + \mu^2(y)/\sigma^2(y))dy} du\right) dz \\ & \leq \int_{\epsilon_0 \wedge x}^x \exp\left(\int_z^{x^*} \frac{\mu^2(u)/\sigma^2(u)}{\int_0^{\epsilon_0 \wedge x} (2\delta + \mu^2(y)/\sigma^2(y))dy} du\right) dz \end{aligned}$$

is finite for $x \in [0, x^*]$. This completes the proof. $\square$

A.3. Proof of Lemma 2

(i) Similar to [24], we prove these inequalities by using proof by contradiction. Suppose there exists some $x \geq x^*$ such that $g'_1(x) \leq 0$ ($g'_2(x) < 0$). Note $g'_1(x^*) = 1$ ($g'_2(x^*) = 0$). Then if we define $x_0 = \inf\{x \geq x^*: g'_1(x) \leq 0\}$ ($x_0 = \inf\{x \geq x^*: g'_2(x) < 0\}$) then x_0 is finite and greater than x^* , and furthermore,

$$g'_1(x) > 0 \quad (g'_2(x) \geq 0), \quad x \in [x^*, x_0], \quad (\text{A9})$$

$$g'_1(x_0) = 0 \quad (g'_2(x_0) = 0), \quad (\text{A10})$$

$$g'_1(x) < 0 \quad (g'_2(x) < 0), \quad \text{for } x_0 < x < x_0 + \delta \text{ and some } \delta > 0. \quad (\text{A11})$$

Equation (A9) along with $g_1(x^*) = 0$ ($g_2(x^*) = 1$) implies that

$$g_1(x_0) > 0 \quad (g_2(x_0) > 0). \quad (\text{A12})$$

Equations (A9)–(A11) together imply $g''_1(x_0) < 0$ ($g''_2(x_0) < 0$). However, from Notation 3 we know, for $i = 1, 2$, $\frac{1}{2}\sigma^2(x_0)g''_i(x_0) + \mu(x_0)g'_i(x_0) - \delta g_i(x_0) = 0$ and, thus,

$$g''_i(x_0) = \frac{2}{\sigma^2(x_0)}(-\mu(x_0)g'_i(x_0) + \delta g_i(x_0)) = \frac{2}{\sigma^2(x_0)}\delta g_i(x_0) > 0, \quad i = 1, 2,$$

where the last equality follows from $g'_i(x_0) = 0$ (see (A10)) and the last inequality follows from noting $g_i(x_0) > 0$ (see (A12)). Thus we have arrived at a contradiction.

(ii) From the expressions for $C_2(b)$, $C_4(b)$, and $C_5(b)$ in (37)–(39) we can see that all three quantities are strictly positive since $g'_1(x) > 0$ and $g'_2(x) \geq 0$ for $x \geq x^*$.

A.4. Proof of Theorem 2

(i)–(iii) From the definitions $C_2(\cdot)$, $C_4(\cdot)$, and $C_5(\cdot)$ in (37)–(38) we observe that the following hold

$$\begin{aligned} v_b(x^* -) &= C_2(b) \int_0^{x^*} e^{\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy} dz = C_4(b)g_1(x^*) + C_5(b)g_2(x^*) = v_b(x^* +), \\ v'_b(x^* -) &= C_2(b) = C_4(b)g'_1(x^*) + C_5(b)g'_2(x^*) = v'_b(x^* +), \\ v'_b(b -) &= v'_b(b +) = 1. \end{aligned} \quad (\text{A13})$$

By using the above equations and the expression of v_b in (45) we can verify that v_b is continuously differentiable on $(0, +\infty)$, and twice continuously differentiable on each of the intervals, $(0, x^*)$, (x^*, b^*) , and $(b^*, +\infty)$.

It has been shown in Lemma 2 that $C_2(b)$, $C_4(\cdot)$, and $C_5(\cdot)$ are all strictly positive, and $g'_1(x) > 0$ and $g'_2(x) \geq 0$ for $x \geq x^*$. We can directly verify from (45) that

$$v'_b(x) = \begin{cases} C_2(b)e^{\int_x^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy} > 0, & x \in (0, x^*], \\ C_4(b)g'_1(x) + C_5(b)g'_2(x) > 0, & x \in (x^*, b], \end{cases} \quad (\text{A14})$$

$$v''_b(x) = -C_2(b) \frac{\mu(x)}{\sigma^2(x)A(x)} e^{\int_x^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy} < 0, \quad x \in (0, x^*], \quad (\text{A15})$$

where the last inequality follows from noting $A(x) > 0$ for $0 < x < x^*$ (see Lemma 1), $\mu(x) > 0$, and $C_2(b) > 0$ (2). It follows from (46) that for $0 < x < x^*$,

$$\arg \max_a \mathcal{L}_{v_b}(a, x) = \arg \max_a \left(\frac{1}{2} \sigma^2(x) a^2 v''_b(x) + \mu(x) a v'_b(x) - \delta v_b(x) \right) = -\frac{\mu(x)}{\sigma^2(x)} \frac{v'_b(x)}{v''_b(x)} = A(x), \quad (\text{A16})$$

where the second to the last equality follows from noting $v''_b(x) < 0$ (see (A15)) and the last equality follows from using the expression for v_b in (45). Since it has been shown in Lemma 1 that $0 \leq A(x) \leq 1$ for $0 \leq x \leq x^*$, we can conclude from (A16) that $\arg \max_{a \in [0, 1]} \mathcal{L}_{v_b}(a, x) = A(x)$ for $x \in (0, x^*)$ and, hence,

$$\max_a \mathcal{L}_{v_b}(a, x) = \mathcal{L}_{v_b}(A(x), x) = \frac{1}{2} \sigma^2(x) A^2(x) v''_b(x) + \mu(x) A(x) v'_b(x) - \delta v_b(x), \quad x \in (0, x^*). \quad (\text{A17})$$

By plugging (45), (A14), and (A15) into the expressions for $\mathcal{L}_{v_b}(A(x), x)$, we get

$$\mathcal{L}_{v_b}(A(x), x) = \frac{1}{2} \sigma^2(x) A^2(x) v''_b(x) + \mu(x) A(x) v'_b(x) - \delta v_b(x) = 0, \quad x \in (0, x^*), \quad (\text{A18})$$

and, thus,

$$\max_a \mathcal{L}_{v_b}(a, x) = \mathcal{L}_{v_b}(A(x), x) = 0, \quad x \in (0, x^*). \quad (\text{A19})$$

By noting that g_1 and g_2 is a set of fundamental solution to $\frac{1}{2} \sigma^2(x) g''(x) + \mu(x) g'(x) - \delta g(x) = 0$, we know v_b is a solution to the same equation. That is,

$$\mathcal{L}_{v_b}(1, x) = \frac{1}{2} \sigma^2(x) v''_b(x) + \mu(x) v'_b(x) - \delta v_b(x) = 0, \quad x \in (x^*, b). \quad (\text{A20})$$

Thus,

$$v''_b(x^*+) = \lim_{x \downarrow x^*} v''_b(x) = \frac{2}{\sigma^2(x^*)} (-\mu(x^*) v'_b(x^*) + \delta v_b(x^*)). \quad (\text{A21})$$

By noting $A(x^*) = 1$ (see Lemma 1) and (A18) we have

$$\begin{aligned} v''_b(x^*-) &= \lim_{x \uparrow x^*} v''_b(x) = \lim_{x \uparrow x^*} \left(\frac{2}{\sigma^2(x)} (-\mu(x) v'_b(x) + \delta v_b(x)) \right) = \frac{2}{\sigma^2(x^*)} (-\mu(x^*) v'_b(x^*) + \delta v_b(x^*)) \\ &= v''_b(x^*+), \end{aligned} \quad (\text{A22})$$

where the last equality follows from (A21). Thus, v_b is twice differentiable at $x = x^*$, and $v_b''(x^*) = v_b''(x^* -) < 0$, where the last inequality is due to (A15). As a result, (A18) holds for $x = x^*$.

(iv) We now proceed to show that $v''(x) \leq 0$ for $x \in (x^*, b)$. We use proof by contradiction. Suppose there exists some $x \in (x^*, b)$ such that $v''(x) > 0$. Note $v_b''(b -) \leq 0$ and $v_b''(x^*) < 0$ (see (ii)). Thus, by the continuity of v_b we know there exist two points x_1 and x_2 with $x^* < x_1 < x_2 \leq b$ such that

$$v_b''(x_1) = 0, \quad v_b''(x_2) = 0, \quad \text{and} \quad v_b''(x) > 0 \quad \text{for } x \in (x_1, x_2). \quad (\text{A23})$$

Starting from the above equation we use $v_b''(x_2)$ represents $v_b''(x_2 -)$ if $x_2 = b$. As a result,

$$v_b''(x) - v_b''(x_i) > 0, \quad x \in (x_1, x_2), \quad i = 1, 2, \quad (\text{A24})$$

and, hence,

$$v_b'(x_2) > v_b'(x_1) > 0, \quad (\text{A25})$$

where the last inequality was shown in (i). It follows from (A20) that

$$v_b''(x) = \frac{2}{\sigma^2(x)} (-\mu(x)v_b'(x) + \delta v_b(x)), \quad x \in [x_1, x_2]. \quad (\text{A26})$$

Hence,

$$-\mu(x)v_b'(x) + \delta v_b(x) > 0, \quad x \in (x_1, x_2) \quad \text{and} \quad -\mu(x_i)v_b'(x_i) + \delta v_b(x_i) = 0, \quad i = 1, 2. \quad (\text{A27})$$

As a result,

$$(-\mu(x)v_b'(x) + \delta v_b(x)) - (-\mu(x_i)v_b'(x_i) + \delta v_b(x_i)) > 0, \quad x \in (x_1, x_2), \quad i = 1, 2. \quad (\text{A28})$$

Dividing both sides of the above equation by $x - x_i$ and then letting $x \downarrow x_1$ and $x \uparrow x_2$ yield

$$(\delta - \mu'(x_1))v_b'(x_1) - \mu(x_1)v_b''(x_1) \geq 0 \quad \text{and} \quad (\delta - \mu'(x_2))v_b'(x_2) - \mu(x_2)v_b''(x_2) \leq 0, \quad (\text{A29})$$

which combined with (A23) implies $(\delta - \mu'(x_1))v_b'(x_1) \geq 0 \geq (\delta - \mu'(x_2))v_b'(x_2)$. By noting $v_b'(x_i) > 0$ for $i = 1, 2$ (see (A25)), we can see that the above equation is impossible because $\mu'(x) < \delta$ for all $x > 0$. Thus, we have now shown the concavity of v_b on $[x^*, b]$.

For any real number a , define

$$h_a(x) = \frac{\partial \mathcal{L}_{v_b}(a, x)}{\partial a}, \quad x \in [x^*, b]. \quad (\text{A30})$$

Then, for $x \in [x^*, b)$ and $a \geq 0$,

$$h_a(x) = a\sigma^2(x)v_b''(x) + \mu(x)v_b'(x) \geq \sigma^2(x)v_b''(x) + \mu(x)v_b'(x) = h_{a=1}(x), \quad (\text{A31})$$

where the last inequality follows from noting $v_b''(x) \leq 0$ for $x \in [x^*, b)$. From (A20) it follows that

$$\sigma^2(x)v_b''(x) = 2(\delta v_b(x) - \mu(x)v_b'(x)), \quad x \in [x^*, b). \quad (\text{A32})$$

Hence, $h_{a=1}(x) = \sigma^2(x)v_b''(x) + \mu(x)v_b'(x) = 2(\delta v_b(x) - \mu(x)v_b'(x)) + \mu(x)v_b'(x) = 2\delta v_b(x) - \mu(x)v_b'(x)$ for $x \in [x^*, b)$. Therefore, $h'_{a=1}(x) = (2\delta - \mu'(x))v_b'(x) - \mu(x)v_b''(x)$ for $x \in [x^*, b)$.

By noting $\mu(x) \geq 0$, $\mu'(x) \leq 2\delta$, and $v_b'(x) > 0$ for $x \geq 0$, and $v_b''(x) \leq 0$ for $x \in [x^*, b)$ (as has been proven in the first half of the proof of (ii)), it follows that $h'_{a=1}(x) \geq 0$, $x \in [x^*, b)$. As a result,

$$h_{a=1}(x) \geq h_1(x^*) = \left. \frac{\partial \mathcal{L}_{v_b}(a, x^*)}{\partial a} \right|_{a=1}, \quad x \in [x^*, b), \quad (\text{A33})$$

where the last equality follows from (A30). Direct calculation based on (46) and (45) shows

$$\left. \frac{\partial \mathcal{L}_{v_b}(a, x^*)}{\partial a} \right|_{a=A(x^*)} = 0.$$

Since $A(x^*) = 1$ (see (29)), we have

$$\left. \frac{\partial \mathcal{L}_{v_b}(a, x^*)}{\partial a} \right|_{a=1} = \left. \frac{\partial \mathcal{L}_{v_b}(a, x^*)}{\partial a} \right|_{a=A(x^*)} = 0. \quad (\text{A34})$$

By combining (A30), (A31), and (A34) we conclude

$$\frac{\partial \mathcal{L}_{v_b}(a, x)}{\partial a} = h_a(x) \geq h_{a=1}(x) = 0, \quad x \in [x^*, b)$$

for $a \geq 0$. This implies $\max_{a \in [0, 1]} \mathcal{L}_{v_b}(a, x) = \mathcal{L}_{v_b}(1, x)$ for $x \in [x^*, b)$, which along with (A20) implies $\max_{a \in [0, 1]} \mathcal{L}_{v_b}(a, x) = \mathcal{L}_{v_b}(1, x) = 0$ for $x \in (x^*, b)$.

A.5. Proof of Lemma 3

Recall that v_b defined in (45) is twice continuously differentiable (Theorem 2(i)). Thus, $v_b''(x^*+) = v_b''(x^*-)$, which by direct calculation using (45) leads to

$$C_4(b)g_1''(x^*) + C_5(b)g_2''(x^*) = -C_2(b) \frac{\mu(x^*)}{\sigma^2(x^*)A(x^*)}. \quad (\text{A35})$$

Now plugging in (37) and (38) and then getting rid of the common denominator on both sides of the newly obtained equation, we obtain

$$g_1''(x^*) + g_2''(x^*) \int_0^{x^*} e^{\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy} dz = -\frac{\mu(x^*)}{\sigma^2(x^*)A(x^*)}. \quad (\text{A36})$$

It follows from (47), (38), to (37) that

$$\xi(x^*) = g_1''(x^*) + g_2''(x^*) \int_0^{x^*} e^{\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy} dz = -\frac{\mu(x^*)}{\sigma^2(x^*)A(x^*)} < 0, \quad (\text{A37})$$

where the last inequality follows from noting $\mu(x^*) > \mu(0) > 0$ and $A(x^*) = 1$. The inequality (A37) along with the definition for b^* in (48) implies that $b^* > x^*$.

A.6. Proof of Lemma 4

All the results in (i)–(iii) directly follow from Theorem 2 except for the second-order differentiability of $v_{b^*}(x)$ at $x = b^*$. Note $b^* > x^*$ (see Lemma 3) and $b^* < +\infty$. Then by noting $v_b''(b-) = \xi(b)$, Notation 4, and the continuity of $\xi(b)$ with respect to b , we have $v_{b^*}''(b^*-) = 0$. Since from (45) we know $v_{b^*}''(b^*+) = 0$, we conclude $v_{b^*}''(b^*)$ exists and $v_{b^*}''(b^*) = 0$.

(iv) It follows from Theorem 2(iv) that $\sup_{a \in [0, 1]} \mathcal{L}_{v_{b^*}}(a, x) = \mathcal{L}_{v_{b^*}}(1, x)$ for $x > b^*$. Note from (45) and (46) we can obtain $\mathcal{L}_{v_{b^*}}(1, x) = \mu(x) - \delta(v_{b^*}(b^*) + x - b^*)$ for $x > b^*$ and, thus,

$$\frac{\partial}{\partial x} \mathcal{L}_{v_{b^*}}(1, x) = \mu'(x) - \delta \leq 0 \quad \text{for } x > b^*.$$

Consequently, $\mathcal{L}_{v_{b^*}}(1, x) \leq \mathcal{L}_{v_{b^*}}(1, b^*) = 0$ for $x > b^*$, where the last equality follows from Theorem 2(iii).

A.7. Proof of Theorem 4

From Theorem 4 we know that $\max_{a \in [0, 1]} \mathcal{L}_v(a, x) = \mathcal{L}_v(A(x), x) = 0$ for $0 \leq x \leq x^*$ and $\sup_{a \in [0, 1]} \mathcal{L}_v(a, x) = \mathcal{L}_v(1, x) \leq 0$ for $x \geq x^*$. Moreover, it follows from Theorem 4(i) that $v_{b^*}''(b^*) = 0$. Therefore, it follows from Theorem 2(ii) and (iii) that $v_{b^*}''(x) \leq 0$ for $0 \leq x \leq b^*$. Note that from Theorem 2(i) we know $v_{b^*}'(b^*) = 1$. Consequently, $v_{b^*}'(x) \geq v_{b^*}'(b^*) = 1$ for $0 \leq x \leq b^*$. Further note from (45) that $v_{b^*}(0) = 0$ and from Theorem 4 that v_{b^*} is twice continuously differentiable. Thus, v_{b^*} is a function that fulfills all the conditions in Theorem 1 and, as a result,

$$v_{b^*}(x) \geq V(x), \quad x \geq 0. \quad (\text{A38})$$

It follows from Theorem 3 that $v_{b^*}(x) = \mathcal{P}(\pi^*, L^{b^*})(x)$ for $x \geq 0$. Thus, $v_{b^*}(x) \leq \sup_{(\pi, L) \in \mathcal{A}} \mathcal{P}(\pi, L)(x) = V(x)$ for $x \geq 0$, which along with (A38) implies $v_{b^*}(x) = V(x)$. As a result, $V(x) = \mathcal{P}(\pi^*, L^{b^*})(x)$ for $x \geq 0$. This completes the proof.

A.8. Proof of Lemma 5

Since $b^* = +\infty$, according to Notation 4 we have $v_b''(b-) = C_4(b)g_1''(b) + C_5(b)g_2''(b) < 0$ for all $b > x^*$. Therefore, for any $b > x^*$, all the desired results follow immediately by Theorem 2.

A.9. Proof of Lemma 6

Since $b^* = +\infty$, as explained in the discussion surrounding (55), we have

$$g_1''(b) < -g_2''(b) \int_0^{x^*} e^{\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy} dz \quad \text{for all } b > x^*,$$

where the last inequality follows from noting $g_2''(b) > 0$ for all $b > x^*$ (54). Hence, $g_1'(b)$ is decreasing on $[x^*, +\infty)$. Recall that $g_1'(b) > 0$ on $[x^*, +\infty)$ (Lemma 2). Therefore, $g_1'(b)$ converges to a non-negative constant as $b \rightarrow +\infty$. From (54) and Lemma 2 we can obtain that $g_2'(b)$ is non-negative and increasing on $[x^*, +\infty)$. Hence, $g_2'(b)$ converges to a non-negative constant or $+\infty$. As a result,

$$g_1'(b) + g_2'(b) \int_0^{x^*} e^{\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy} dz$$

converges to a non-negative constant or $+\infty$. Consequently, by noticing the expressions for $C_i(b)$ for $i = 2, 4, 5$ in (37) and (38), we can obtain $\lim_{b \uparrow +\infty} C_2(b)$, $\lim_{b \uparrow +\infty} C_4(b)$, and $\lim_{b \uparrow +\infty} C_5(b)$ all exist and are non-negative finite constants, and $\lim_{b \uparrow +\infty} C_2(b)$ exists and is a non-positive constant.

A.10. Proof of Theorem 5

(i) Recall from Lemma 6 that in the case $b^* = +\infty$, $C_2(b)$, $C_4(b)$, and $C_5(b)$ all converge to finite constants as b converges to $+\infty$. Consequently, we can directly observe from (45) that for any $x > 0$, as b increases, $v_b(x)$ converges to

$$\lim_{b \rightarrow +\infty} v_b(x) = \begin{cases} \left(\lim_{b \rightarrow +\infty} C_2(b) \right) \int_0^x e^{\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy} dz, & x \in [0, x^*), \\ \left(\lim_{b \rightarrow +\infty} C_4(b) \right) g_1(x) + \left(\lim_{b \rightarrow +\infty} C_5(b) \right) g_2(x), & x \in [x^*, \infty). \end{cases} \quad (\text{A39})$$

Define

$$\bar{v}(x) = \lim_{b \rightarrow +\infty} v_b(x), \quad x \geq 0. \quad (\text{A40})$$

Since $v_b(x) = \mathcal{P}(\pi^*, L_b)(x) \leq \sup_{(\pi, L) \in \mathcal{A}} \mathcal{P}(\pi, L)(x) = V(x)$, it is sufficient to prove that $\bar{v}(x) \geq V(x)$ for $x \geq 0$. From (A39)–(A40) we can observe that $\bar{v}$ is twice continuously differentiable, $\bar{v}'(x) = \lim_{b \rightarrow +\infty} v'_b(x)$ and $\bar{v}''(x) = \lim_{b \rightarrow +\infty} v''_b(x)$ for $x \geq 0$. Therefore, from the definition of $\mathcal{L}_g$, we can obtain

$$\mathcal{L}_{\bar{v}}(a, x) = \lim_{b \rightarrow +\infty} \mathcal{L}_{v_b}(a, x) \quad \text{for any real } a \text{ and } x \geq 0. \quad (\text{A41})$$

Recall from Lemma 5 that for any $b > x^*$ the following hold:

$$\max_{a \in [0, 1]} \mathcal{L}_{v_b}(a, x) = \mathcal{L}_{v_b}(A(x), x) = 0, \quad x \in [0, x^*), \quad (\text{A42})$$

$$\max_{a \in [0, 1]} \mathcal{L}_{v_b}(a, x) = \mathcal{L}_{v_b}(1, x) = 0, \quad x \in (x^*, b). \quad (\text{A43})$$

By combining (A41)–(A43) we arrive at

$$\max_{a \in [0, 1]} \mathcal{L}_{\bar{v}}(a, x) = \mathcal{L}_{\bar{v}}(A(x), x) = 0, \quad x \in [0, x^*], \quad (\text{A44})$$

$$\max_{a \in [0, 1]} \mathcal{L}_{\bar{v}}(a, x) = \mathcal{L}_{\bar{v}}(1, x) = 0, \quad x \in (x^*, \infty). \quad (\text{A45})$$

Since from Lemma 5 it follows that for $b > x^*$, $v'_b(x) \geq 1$ for $x \geq 0$. Thus,

$$\bar{v}'(x) = \lim_{b \rightarrow +\infty} v'_b(x) \geq 1, \quad x \geq 0. \quad (\text{A46})$$

Note that obviously $\bar{v}(0) = \lim_{b \rightarrow +\infty} v_b(0) = 0$. We have thus shown that all the conditions in Theorem 1 are satisfied by the function $\bar{v}(x)$ and as a result, $\bar{v}(x) \geq V(x)$ for $x \geq 0$. On the other hand, $\bar{v}(x) = \lim_{b \rightarrow +\infty} v_b(x) = \lim_{b \rightarrow +\infty} \mathcal{P}(\pi^*, L^b)(x) \leq V(x)$ where the last equality follows from Theorem 3. Thus, we can conclude $\bar{v}(x) = \lim_{b \rightarrow +\infty} v_b(x)$ for $x \geq 0$.

Since we have shown that $\bar{v}$ is the value function and that $\mathcal{L}_{\bar{v}}(A(x), x) = \max_{a \in [0, 1]} \mathcal{L}_{\bar{v}}(a, x)$ for $0 \leq x \leq x^*$ and $\mathcal{L}_{\bar{v}}(1, x) = \max_{a \in [0, 1]} \mathcal{L}_{\bar{v}}(a, x)$ for $x > x^*$, the strategy stated in (ii) is optimal.

(ii) We first show that $\bar{v}'(x) > 1$ for all $x > 0$ using proof of contradiction. Suppose this is not true. Then by noting $\bar{v}'(x) \geq 1$ (A46), we know there exists a finite $\bar{x} > 0$ such that $\bar{v}'(\bar{x}) = 1$.

Since v_b is concave and so as its limit $\bar{v}''(\bar{x}) \leq 0$. Therefore,

$$\bar{v}'(x) = 1, \quad x \geq \bar{x}, \quad (\text{A47})$$

and, hence,

$$\bar{v}''(\bar{x}) = 0. \quad (\text{A48})$$

From the construction of v_b we can easily verify that v_b is a twice continuously differentiable solution on $(0, b)$ to the following boundary value equations:

$$\frac{1}{2}\sigma^2(x)A(x)g''(x) + \mu(x)A(x)g'(x) - \delta g(x) = 0, \quad x \in (0, x^*], \quad (\text{A49})$$

$$\frac{1}{2}\sigma^2(x)g''(x) + \mu(x)g'(x) - \delta g(x) = 0, \quad x \in (x^*, b), \quad (\text{A50})$$

$$g(0) = 0, \quad g'(b) = 1. \quad (\text{A51})$$

Since we have already shown in (i) that $\bar{v}(x) = \lim_{b \rightarrow +\infty} v_b(x)$, $\bar{v}'(x) = \lim_{b \rightarrow +\infty} v'_b(x)$, and $\bar{v}''(x) = \lim_{b \rightarrow +\infty} v''_b(x)$, and thus we can verify that $\bar{v}(x)$ satisfies (A49) and (A50). Note $\bar{v}(0) = \lim_{b \rightarrow +\infty} v_b(0) = 0$ and that we have shown $\bar{v}'(\bar{x}) = 1$. Thus, we can conclude $\bar{v}$ is a solution to the above boundary value equations when b is set to be $\bar{x}$. Using the standard theory in second-order ODEs we know that for any fixed b , the above boundary value equations admit a unique twice differentiable solution on $[0, b]$. Hence, the two solutions to the above boundary value equations, $\bar{v}(x)$ and $v_{\bar{x}}(x)$, are identical on $[0, \bar{x}]$, that is, $\bar{v}(x) = v_{\bar{x}}(x)$ for $x \in [0, \bar{x}]$. Since $\bar{v}'(x) \equiv 1$ and $\bar{v}'(x) \equiv 1$ for $x \geq \bar{x}$, we can obtain that for $x > \bar{x}$, $\bar{v}(x) = \bar{v}(\bar{x}) + (x - \bar{x}) = v_{\bar{x}}(\bar{x}) + (x - \bar{x}) = v_{\bar{x}}(x)$ for $x \geq 0$. Hence, $v''_{\bar{x}}(\bar{x} -) = \bar{v}''(\bar{x} -) = 0$, where the last equality follows from (A48). Then from the comment right below Notation 4, we can obtain $b^* = \bar{x}$, which violates the underlying assumption that we are considering the case $b^* = +\infty$.

Now we have shown that $\bar{v}'(x) > 1$ for all $x > 0$. We use proof by contradiction again to show that no optimal dividend strategy exists. Suppose this is not true, that is, there exists an admissible strategy $(\bar{\pi}, \bar{L})$ such that $V(\bar{\pi}, \bar{L})(x) = V(x)$ for $x \geq 0$. For any $t > 0$, by applying the Itô's formula to $e^{-\delta(\tau^{\bar{\pi}, \bar{L}} \wedge t)} \bar{v}(X_{\tau^{\bar{\pi}, \bar{L}} \wedge t}^{\bar{\pi}, \bar{L}})$ we can obtain

$$\begin{aligned} & E_x \left[e^{-\delta(\tau^{\bar{\pi}, \bar{L}} \wedge t)} \bar{v}(X_{\tau^{\bar{\pi}, \bar{L}} \wedge t}^{\bar{\pi}, \bar{L}}) - \bar{v}(X_{0-}) \right] \\ &= E_x \left[\int_0^{\tau^{\bar{\pi}, \bar{L}} \wedge t} e^{-\delta s} \left(\frac{1}{2} \sigma^2(X_{s-}^{\bar{\pi}, \bar{L}}) (\bar{\pi}_{s-})^2 \bar{v}''(X_{s-}^{\bar{\pi}, \bar{L}}) + \mu(X_{s-}^{\bar{\pi}, \bar{L}}) \bar{\pi}_{s-} \bar{v}'(X_{s-}^{\bar{\pi}, \bar{L}}) - \delta \bar{v}(X_{s-}^{\bar{\pi}, \bar{L}}) \right) ds \right. \\ &\quad \left. + \int_0^{\tau^{\bar{\pi}, \bar{L}} \wedge t} e^{-\delta s} \sigma(X_{s-}^{\bar{\pi}, \bar{L}}) \bar{\pi}_{s-} \bar{v}(X_{s-}^{\bar{\pi}, \bar{L}}) dW_s - \int_0^{\tau^{\bar{\pi}, \bar{L}} \wedge t} e^{-\delta s} \bar{v}'(X_{s-}^{\bar{\pi}, \bar{L}}) d\bar{L}_s \right. \\ &\quad \left. + \sum_{0 \leq s \leq \tau^{\bar{\pi}, \bar{L}} \wedge t} e^{-\delta s} (\bar{v}(X_s^{\bar{\pi}, \bar{L}}) - \bar{v}(X_{s-}^{\bar{\pi}, \bar{L}})) \right] \\ &= E_x \left[\int_0^{\tau^{\bar{\pi}, \bar{L}} \wedge t} e^{-\delta s} \mathcal{L}_{\bar{v}}(\bar{\pi}_{s-}, X_{s-}^{\bar{\pi}, \bar{L}}) ds \right. \\ &\quad \left. + \int_0^{\tau^{\bar{\pi}, \bar{L}} \wedge t} e^{-\delta s} \sigma(X_{s-}^{\bar{\pi}, \bar{L}}) \bar{\pi}_{s-} \bar{v}'(X_{s-}^{\bar{\pi}, \bar{L}}) dW_s - \int_0^{\tau^{\bar{\pi}, \bar{L}} \wedge t} e^{-\delta s} \bar{v}'(X_{s-}^{\bar{\pi}, \bar{L}}) d\bar{L}_s \right. \\ &\quad \left. + \sum_{0 \leq s \leq \tau^{\bar{\pi}, \bar{L}} \wedge t} e^{-\delta s} (\bar{v}(X_s^{\bar{\pi}, \bar{L}}) - \bar{v}(X_{s-}^{\bar{\pi}, \bar{L}})) \right], \quad (\text{A52}) \end{aligned}$$

where $\tilde{\bar{L}}$ represents the continuous part of $\bar{L}$. Note by (A44) and (A45) we have $\mathcal{L}_{\bar{v}}(\bar{\pi}_{s-}, X_{s-}^{\bar{\pi}, \bar{L}}) \leq 0$. Note that

$$\int_0^{\tau^{\bar{\pi}, \bar{L}} \wedge t} e^{-\delta s} \sigma(X_{s-}^{\bar{\pi}, \bar{L}}) \bar{\pi}_{s-} f(X_{s-}^{\bar{\pi}, \bar{L}}) dW_s$$

is a local martingale and so there exists a sequence of stopping times, τ_n , such that

$$\int_0^{\tau^{\bar{\pi}, \bar{L}} \wedge t \wedge \tau_n} e^{-\delta s} \sigma(X_{s-}^{\bar{\pi}, \bar{L}}) \bar{\pi}_{s-} f(X_{s-}^{\bar{\pi}, \bar{L}}) dW_s$$

is a martingale. Thus,

$$E_x \left[\int_0^{\tau^{\bar{\pi}, \bar{L}} \wedge t \wedge \tau_n} e^{-\delta s} \sigma(X_{s-}^{\bar{\pi}, \bar{L}}) \bar{\pi}_{s-} f(X_{s-}^{\bar{\pi}, \bar{L}}) dW_s \right] = 0, \quad x \geq 0. \quad (\text{A53})$$

As a result, we arrive at

$$\begin{aligned} & E_x [e^{-\delta(\tau^{\bar{\pi}, \bar{L}} \wedge t \wedge \tau_n)} \bar{v}(X_{\tau^{\bar{\pi}, \bar{L}} \wedge t}^{\bar{\pi}, \bar{L}})] - \bar{v}(x) \\ & \leq -E_x \left[\int_0^{\tau^{\bar{\pi}, \bar{L}} \wedge t \wedge \tau_n} e^{-\delta s} \bar{v}'(X_{s-}^{\bar{\pi}, \bar{L}}) d\tilde{L}_s \right] + E_x \left[\sum_{0 \leq s \leq \tau^{\bar{\pi}, \bar{L}} \wedge t \wedge \tau_n} e^{-\delta s} (\bar{v}(X_s^{\bar{\pi}, \bar{L}}) - \bar{v}(X_{s-}^{\bar{\pi}, \bar{L}})) \right]. \end{aligned} \quad (\text{A54})$$

Note $E_x [e^{-\delta(\tau^{\bar{\pi}, \bar{L}} \wedge t \wedge \tau_n)} \bar{v}(X_{\tau^{\bar{\pi}, \bar{L}} \wedge t}^{\bar{\pi}, \bar{L}})] \geq 0$. Hence,

$$\bar{v}(x) \geq E_x \left[\int_0^{\tau^{\bar{\pi}, \bar{L}} \wedge t \wedge \tau_n} e^{-\delta s} \bar{v}'(X_{s-}^{\bar{\pi}, \bar{L}}) d\tilde{L}_s + \sum_{0 \leq s \leq \tau^{\bar{\pi}, \bar{L}} \wedge t \wedge \tau_n} e^{-\delta s} (\bar{v}(X_s^{\bar{\pi}, \bar{L}}) - \bar{v}(X_{s-}^{\bar{\pi}, \bar{L}})) \right]. \quad (\text{A55})$$

By letting t and n converge $+\infty$ on both sides and using the monotone convergence we obtain

$$\bar{v}(x) \geq E_x \left[\int_0^{\tau^{\bar{\pi}, \bar{L}}} e^{-\delta s} \bar{v}'(X_{s-}^{\bar{\pi}, \bar{L}}) d\tilde{L}_s + \sum_{0 \leq s \leq \tau^{\bar{\pi}, \bar{L}}} e^{-\delta s} (\bar{v}(X_s^{\bar{\pi}, \bar{L}}) - \bar{v}(X_{s-}^{\bar{\pi}, \bar{L}})) \right]. \quad (\text{A56})$$

By letting t and n converge $+\infty$ on both sides and using the monotone convergence we obtain

$$\bar{v}(x) \geq E_x \left[\int_0^{\tau^{\bar{\pi}, \bar{L}}} e^{-\delta s} \bar{v}'(X_{s-}^{\bar{\pi}, \bar{L}}) d\tilde{L}_s + \sum_{0 \leq s \leq \tau^{\bar{\pi}, \bar{L}}} e^{-\delta s} (\bar{v}(X_s^{\bar{\pi}, \bar{L}}) - \bar{v}(X_{s-}^{\bar{\pi}, \bar{L}})) \right] \quad (\text{A57})$$

$$> E_x \left[\int_0^{\tau^{\bar{\pi}, \bar{L}}} e^{-\delta s} d\tilde{L}_s + \sum_{0 \leq s \leq \tau^{\bar{\pi}, \bar{L}}} e^{-\delta s} (\bar{L}_s - \bar{L}_{s-}) \right] \quad (\text{A58})$$

$$= E_x \left[\int_0^{\tau^{\bar{\pi}, \bar{L}}} e^{-\delta s} d\bar{L}_s \right] = \mathcal{P}(\bar{\pi}, \bar{L})(x) > 0, \quad x > 0, \quad (\text{A59})$$

where the last inequality holds strictly because $\bar{v}'(x) > 1$ for all $x > 0$ (as has been shown at the beginning of the proof to (iii)) and

$$\begin{aligned} E_x \left[\int_0^{\tau_{\bar{\pi}, \bar{L}}} e^{-\delta s} d\tilde{L}_s + \sum_{0 \leq s \leq \tau_{\bar{\pi}, \bar{L}}} e^{-\delta s} (X_{s-}^{\bar{\pi}, \bar{L}} - X_s^{\bar{\pi}, \bar{L}}) \right] \\ = E_x \left[\int_0^{\tau_{\bar{\pi}, \bar{L}}} e^{-\delta s} d\tilde{L}_s + \sum_{0 \leq s \leq \tau_{\bar{\pi}, \bar{L}}} e^{-\delta s} (\bar{L}_s - \bar{L}_{s-}) \right] > 0 \end{aligned}$$

(as can be seen from (A58) and (A59)). Note that $\bar{v}(x) = \lim_{b \rightarrow +\infty} v_b(x) \leq V(x)$, which combined with (A59) implies $V(x) \geq \bar{v}(x) > \mathcal{P}(\bar{\pi}, \bar{L})(x)$ for $x > 0$. This contradicts the defining property of $(\bar{\pi}, \bar{L})$: $V(\bar{\pi}, \bar{L})(x) = V(x)$.

A.11. Proof of Theorem 6

Define $f(x) = x$. We can observe that under L^0 , the company pays out all the initial wealth at time 0 and claims bankruptcy. Thus, $\mathcal{P}(\pi^0, L^0)(x) = x = f(x)$ for $x \geq 0$. Therefore, $f(x) = x \leq \sup_{(\pi, L) \in \mathcal{A}} \mathcal{P}(\pi, L)(x) = V(x)$ for $x \geq 0$. Thus, it is sufficient to show $f(x) \geq V(x)$. Note $\mu(0) \leq 0$, $f(x) = x$, $f'(x) = 1$, $f''(x) = 0$, and $\mu'(x) \leq \delta$, we can easily obtain

$$\begin{aligned} \max_{a \in [0, 1]} \left(\frac{1}{2} \sigma^2(x) a^2 f''(x) + \mu(x) a f'(x) - \delta f(x) \right) &= \mu(x) - \delta x \leq \mu(0) + \delta x - \delta x \leq 0, \quad x \geq 0, \\ f'(x) &= 1, \quad x \geq 0. \end{aligned}$$

Further note $f(0) = 0$ and $f(x)$ is sufficiently smooth. Thus, all the conditions in Theorem 1 are fulfilled and, thus, by Theorem 1 we can conclude $f(x) \geq V(x)$. Consequently, $f(x) = V(x)$ and $\mathcal{P}(\pi^0, L^0) = f(x) = V(x)$ for $x \geq 0$.

A.12. Proof of Lemma 7

Recall that we are considering the case $\mu(0) > 0$ and in this case $\mu(x) > 0$ for $0 \leq x < x_0$ and $\mu(x_0) = 0$. Note from (26) $A(0) = 0$ and

$$\lim_{x \rightarrow x_0} A(x) = \lim_{x \rightarrow x_0} \frac{\int_0^x (2\delta + \frac{\mu^2(y)}{\sigma^2(y)}) dy}{\mu(x)} = \infty.$$

Hence, by its continuity we know that $A(x)$ attains 1 for some $x \in (0, x_0)$, which implies that x^* , which is the smallest positive point at which $A(x)$ attains 1, is between 0 and x_0 . From (26) it follows that

$$A(x) \geq \frac{\int_0^x 2\delta dy}{\mu(x)} = \frac{2\delta x}{\mu(x)} \geq \frac{2\delta x}{\mu(0) + \delta x} > 1 \quad \text{for } x > \frac{\mu(0)}{\delta},$$

where the second to the last inequality is due to $\mu'(x) \leq \delta$. Thus, $A(x)$ attains 1 at or before $\mu(0)/\delta$, that is, $x \leq \mu(0)/\delta$.

We can follow the same steps as in the proof of Lemma 1 to show that $A(x)$ is increasing on $[0, x^*]$.

A.13. Proof of Lemma 8

All the steps in the proof of Lemma 3 leading to (A37) remain valid for any x , yielding

$$g_1''(x) + g_2''(x) \int_0^x e^{\int_z^x \frac{\mu(y)}{\sigma^2(y)A(y)} dy} dz = -\frac{\mu(x)}{\sigma^2(x)A(x)}. \quad (\text{A60})$$

Now, note that $x^* < x_0$ (by Lemma 7) and, hence, it follows from (57) that $\mu(x^*) > 0$. Further note that $A(x^*) = 1$ (by Lemma 7). Therefore,

$$g_1''(x^*) + g_2''(x^*) \int_0^{x^*} e^{\int_z^{x^*} \frac{\mu(y)}{\sigma^2(y)A(y)} dy} dz < 0,$$

which, along with the definition for b^* in (48), implies that $b^* > x^*$.

As $\mu(x_0) = 0$, it follows from (A60) that

$$g_1''(x_0) + g_2''(x_0) \int_0^{x_0} e^{\int_z^{x_0} \frac{\mu(y)}{\sigma^2(y)A(y)} dy} dz = -\frac{\mu(x_0)}{\sigma^2(x_0)A(x_0)} = 0,$$

which, along with the definition for b^* in (48), implies that $b^* \leq x_0$.

A.14. Proof of Theorem 7

It follows from Lemma 8 that $\mu(x) > 0$ for $0 \leq x < b^*$. In light of the results in Lemma 7, we observe that the proofs of Theorems 2–4 remain valid and, therefore, these theorems still hold in this case. As a result, all the steps in the proof of Theorem 4 continue to apply. This completes the proof.

A.15. Proof of Equation (83)

Recall that $V^{\alpha, \beta}$ represents the value function in [10] and it has the following representation:

$$V^{\alpha, \beta}(x) = \begin{cases} k_1(\alpha, \beta)(e^{r_+(\alpha)x} - e^{r_-(\alpha)x}), & x \in [0, x_\alpha], \\ \frac{\mu\alpha}{2\gamma}(V^{\alpha, \beta})'(x_\alpha -) \left(\frac{x - x_\alpha + y_\alpha}{y_\alpha} \right)^{1-\Gamma}, & x \in (x_\alpha, x_\beta), \\ k_1(\beta)e^{r_+(\beta)(x-x_1)} + k_2(\beta)e^{r_-(\beta)(x-x_1)}, & x \in [x_\beta, x_1], \\ k_1(\beta) + k_2(\beta) + x - x_1, & x \in [x_1, \infty), \end{cases} \quad (\text{A61})$$

where $\gamma > 0$ is the discount rate,

$$\begin{aligned} r_+(\alpha) &= \frac{-\mu + [\mu^2 + 2\sigma^2\gamma]^{1/2}}{\alpha\sigma^2}, & r_-(\alpha) &= \frac{-\mu - [\mu^2 + 2\sigma^2\gamma]^{1/2}}{\alpha\sigma^2}, \\ r_+(\beta) &= \frac{-\mu + [\mu^2 + 2\sigma^2\gamma]^{1/2}}{\beta\sigma^2}, & r_-(\beta) &= \frac{-\mu - [\mu^2 + 2\sigma^2\gamma]^{1/2}}{\beta\sigma^2}, \\ x_\alpha &= \frac{1}{r_+(\alpha) - r_-(\alpha)} \ln \left(-\frac{r_-(\alpha)}{r_+(\alpha)} \right) > 0, & x_\beta &= \frac{\mu\sigma^2}{\mu^2 + 2\sigma^2\gamma}(\beta - \alpha) + x_\alpha, \\ y_\alpha &= \frac{\mu\sigma^2\alpha}{\mu^2 + 2\sigma^2\gamma}, & \Gamma &= \frac{\mu^2}{\mu^2 + 2\sigma^2\gamma}, \\ x_1 &= x_\beta - \Delta, & \Delta &= \frac{\beta\sigma^2}{2[\mu^2 + 2\sigma^2\gamma]^{1/2}} \log \left(\frac{-\mu + [\mu^2 + 2\sigma^2\gamma]^{1/2}}{\mu + [\mu^2 + 2\sigma^2\gamma]^{1/2}} \right) < 0, \\ k_1(\beta) &= \frac{-r_-(\beta)}{r_+(\beta)(r_+(\beta) - r_-(\beta))}, & k_2(\beta) &= \frac{r_+(\beta)}{r_-(\beta)(r_+(\beta) - r_-(\beta))}, \end{aligned} \quad (\text{A62})$$

and

$$k_1(\alpha, \beta) = \frac{k_1(\beta)r_+(\beta)e^{r_+(\beta)\Delta} + k_2(\beta)r_-(\beta)e^{r_-(\beta)\Delta}}{\frac{\alpha\mu}{2\gamma y_\alpha}(r_+(\alpha)e^{r_+(\alpha)x_\alpha} - r_-(\alpha)e^{r_-(\alpha)x_\alpha})\left(\frac{\beta}{\alpha}\right)^{-\Gamma}(1-\Gamma)}. \quad (\text{A63})$$

In the original form of the formulas for $V^{\alpha,\beta}$, the formulas for the second piece use $(V^{\alpha,\beta})'(x_\alpha)$, which we think should be $(V^{\alpha,\beta})'(x_\alpha -)$. The two quantities are eventually same as $V^{\alpha,\beta}$ is continuously differentiable at x_α . By direct calculation using the first piece of the formulas for $(V^{\alpha,\beta})'$ we obtain

$$\begin{aligned} \frac{\mu\alpha}{2\gamma}(V^{\alpha,\beta})'(x_\alpha -) &= \frac{\mu\alpha}{2\gamma}k_1(\alpha, \beta)(r_+(\alpha)e^{r_+(\alpha)x_\alpha} - r_-(\alpha)e^{r_-(\alpha)x_\alpha}) \\ &= y_\alpha \left(\frac{\beta}{\alpha}\right)^\Gamma \frac{k_1(\beta)r_+(\beta)e^{r_+(\beta)\Delta} + k_2(\beta)r_-(\beta)e^{r_-(\beta)\Delta}}{(1-\Gamma)}, \end{aligned} \quad (\text{A64})$$

where the last equality follows immediately by plugging in (A63). Therefore, the second piece of formulas in (A61), which is the formula of $V^{\alpha,\beta}(x)$ for $x_\alpha \leq x < x_\beta$, becomes

$$\begin{aligned} V^{\alpha,\beta}(x) &= y_\alpha \left(\frac{\beta}{\alpha}\right)^\Gamma \frac{k_1(\beta)r_+(\beta)e^{r_+(\beta)\Delta} + k_2(\beta)r_-(\beta)e^{r_-(\beta)\Delta}}{(1-\Gamma)} \left(\frac{x - x_\alpha + y_\alpha}{y_\alpha}\right)^{1-\Gamma} \\ &= \frac{\mu\sigma^2\alpha}{\mu^2 + 2\sigma^2\gamma} \left(\frac{\beta}{\alpha}\right)^\Gamma \frac{k_1(\beta)r_+(\beta)e^{r_+(\beta)\Delta} + k_2(\beta)r_-(\beta)e^{r_-(\beta)\Delta}}{(1-\Gamma)} \left(\frac{x - x_\alpha + \frac{\mu\sigma^2\alpha}{\mu^2 + 2\sigma^2\gamma}}{\frac{\mu\sigma^2\alpha}{\mu^2 + 2\sigma^2\gamma}}\right)^{1-\Gamma} \\ &= \left(k_1(\beta)r_+(\beta)e^{r_+(\beta)\Delta} + k_2(\beta)r_-(\beta)e^{r_-(\beta)\Delta}\right) \frac{\mu\beta^\Gamma}{2\gamma} \left(\frac{x - x_\alpha + \frac{\mu\sigma^2\alpha}{\mu^2 + 2\sigma^2\gamma}}{\frac{\mu\sigma^2}{\mu^2 + 2\sigma^2\gamma}}\right)^{1-\Gamma}. \end{aligned} \quad (\text{A65})$$

When $\alpha = 0$, $\beta = 1$, by noting $\gamma = \delta$ (because γ and δ represent the same thing, the discount rate, in the reference [10] and this paper, respectively), from the above equations we can conclude $x_\alpha = 0$, $x_\beta = \mu\sigma^2/(\mu^2 + 2\sigma^2\delta) = x^*$, $y_\alpha = 0$, $r_+(1) = \theta_1$, and $r_-(1) = \theta_2$. Using these and noting $\gamma = \delta$ again, we can obtain that when $\alpha = 0$ and $\beta = 1$,

$$\begin{aligned} k_1(1) &= \frac{-\theta_2}{\theta_1(\theta_1 - \theta_2)}, \quad k_2(1) = \frac{\theta_1}{\theta_2(\theta_1 - \theta_2)}, \\ x_1 &= x_\beta - \Delta = x_\beta - \frac{\sigma^2}{2[\mu^2 + 2\sigma^2\delta]^{1/2}} \ln \left(\frac{-\mu + [\mu^2 + 2\sigma^2\delta]^{1/2}}{\mu + [\mu^2 + 2\sigma^2\delta]^{1/2}} \right) \\ &= x^* + \frac{\ln \left(-\frac{\theta_2}{\theta_1} \right)}{\theta_1 - \theta_2} = x^* + \frac{\ln \frac{\theta_2^2(1-\theta_1\frac{\mu}{2\delta})}{\theta_1^2(1-\theta_2\frac{\mu}{2\delta})}}{\theta_1 - \theta_2} = b^*, \end{aligned} \quad (\text{A66})$$

where the last equality follows from (79). Recall $\Gamma = \mu^2/(\mu^2 + 2\sigma^2\gamma)$ and $\gamma = \delta$. Substituting these, (A65), (A66), $x_{\beta=1} = x^*$, and $x_1 = b^*$ into (A61) we have

$$V^{\alpha=0,\beta=1}(x) = \begin{cases} \left(\frac{-\theta_2}{\theta_1-\theta_2}e^{\theta_1\Delta} + \frac{\theta_1}{\theta_1-\theta_2}e^{\theta_2\Delta}\right) \frac{\mu}{2\delta} \left(\frac{x}{x^*}\right)^{1-\Gamma}, & x \in [0, x^*), \\ k_1(1)e^{\theta_1(x-x^*)} + k_2(1)e^{\theta_2(x-x^*)}, & x \in [x^*, b^*), \\ k_1(1) + k_2(1) + x - b^*, & x \in [b^*, \infty). \end{cases}$$

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